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(54) **VOLTAGE REGULATOR MODULE**

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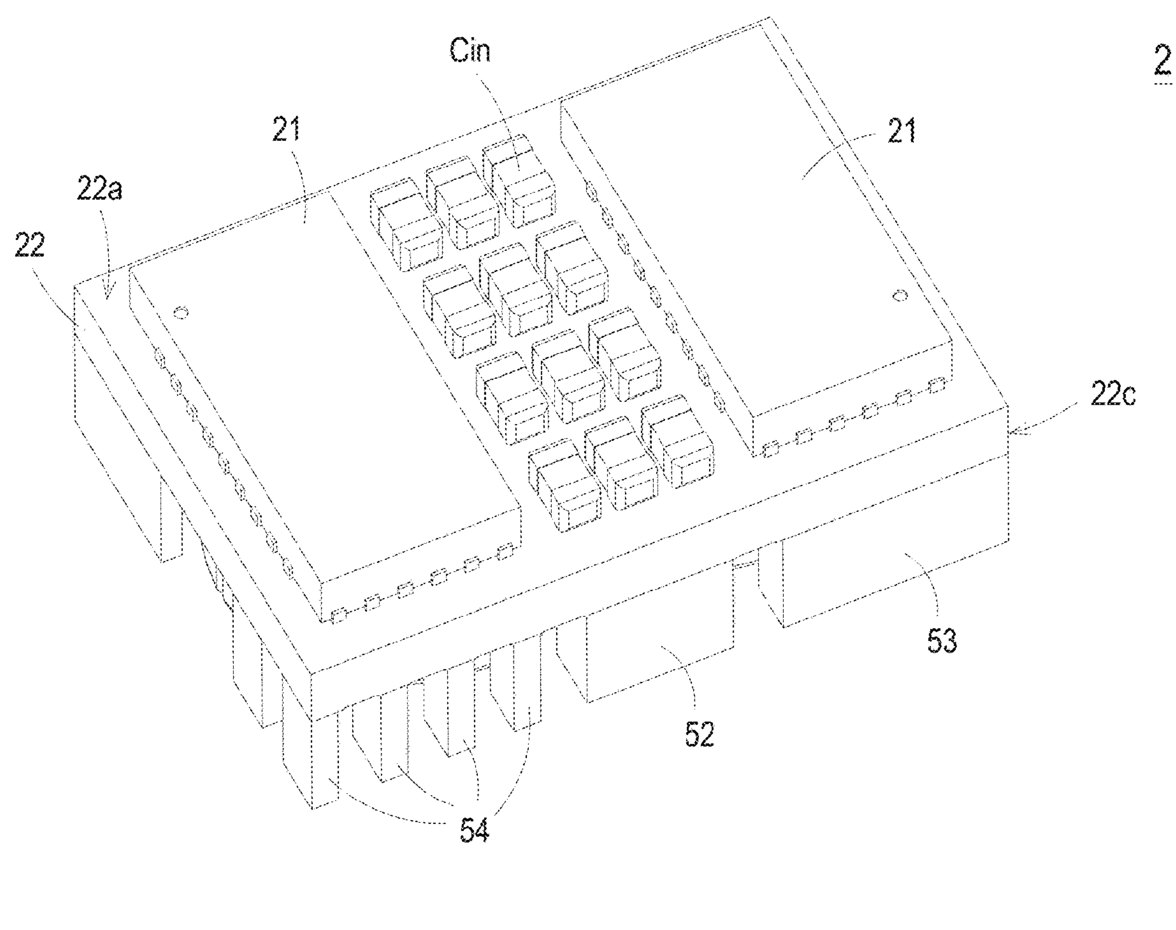
H05K 5/02	(2006.01)
H05K 5/06	(2006.01)

(52) **U.S. Cl.**

CPC ***H02M 3/158*** (2013.01); ***H05K 1/115*** (2013.01); ***H05K 1/181*** (2013.01); ***H05K 1/185*** (2013.01); ***H05K 5/0247*** (2013.01); ***H05K 5/065*** (2013.01); *H02M 1/0048* (2021.05); *H02M 3/003* (2021.05); *H05K 2201/1003* (2013.01); *H05K 2201/10166* (2013.01); *H05K 2201/1031* (2013.01); *H05K 2201/10522* (2013.01)

(57) **ABSTRACT**

A voltage regulator module is provided. The voltage regulator module includes a circuit board assembly and a magnetic core assembly. The circuit board assembly includes a printed circuit board and at least one switch element. The printed circuit board comprises an upper surface, a lower surface, an inner layer and a plurality of conductive portions. The upper surface and the lower surface are opposite to each other. The inner layer is embedded within the printed circuit board. A plurality of conductive portions are electrically connected between the upper surface and the lower surface of the printed circuit board. The at least one switch element is disposed on the upper surface of the printed circuit board. The magnetic core assembly is disposed in the inner layer.



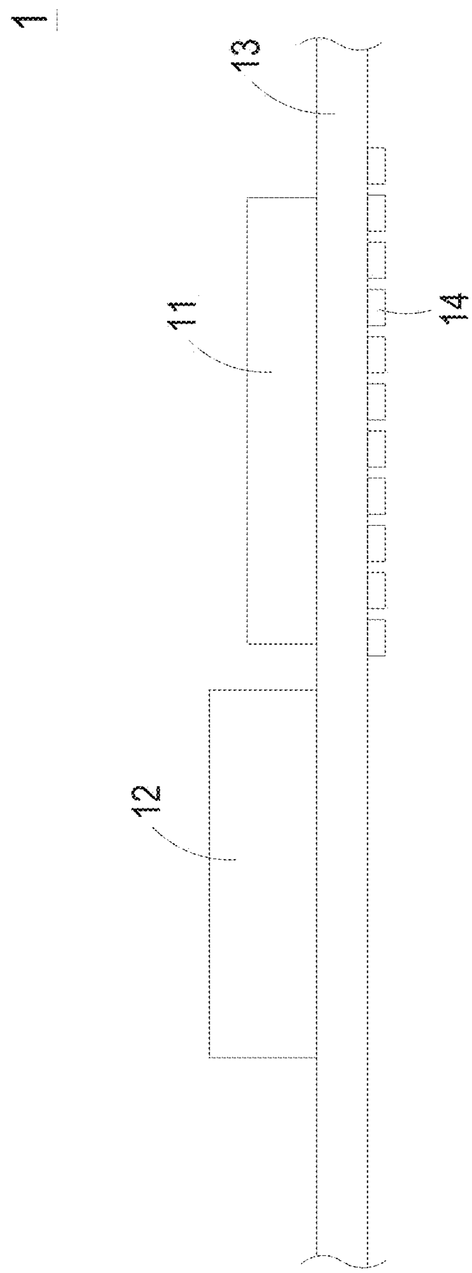


FIG. 1A PRIOR ART

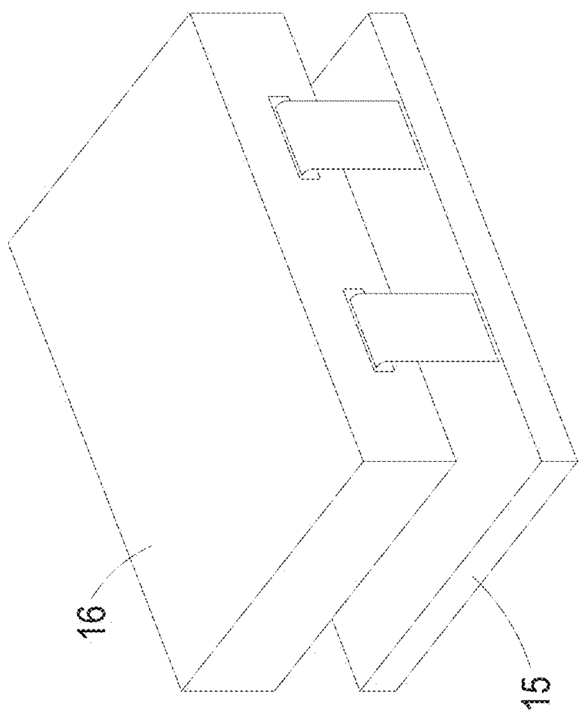


FIG. 1B PRIOR ART

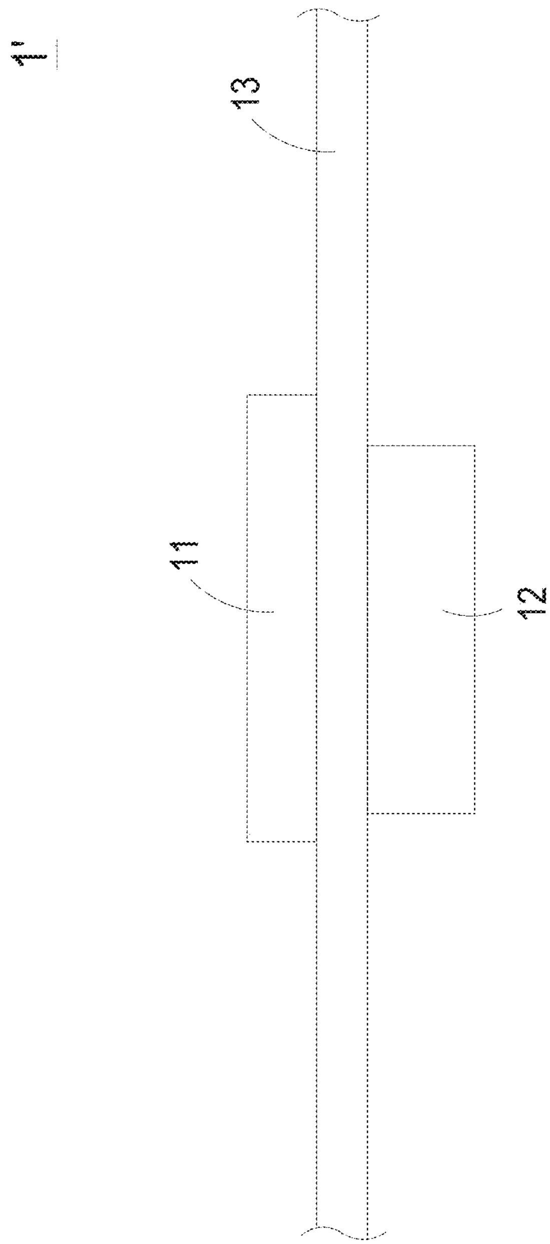


FIG. 2 PRIOR ART

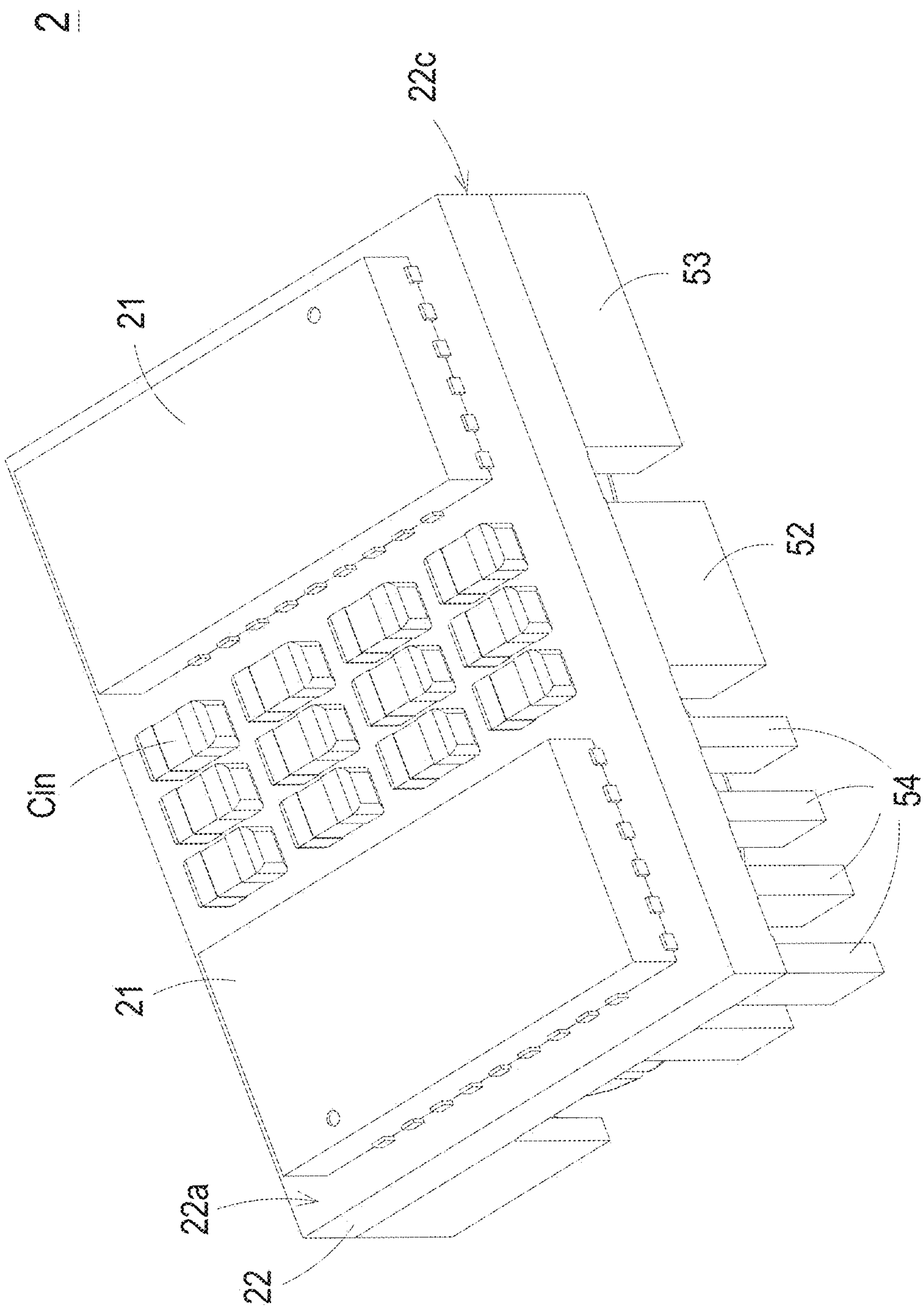


FIG. 3A

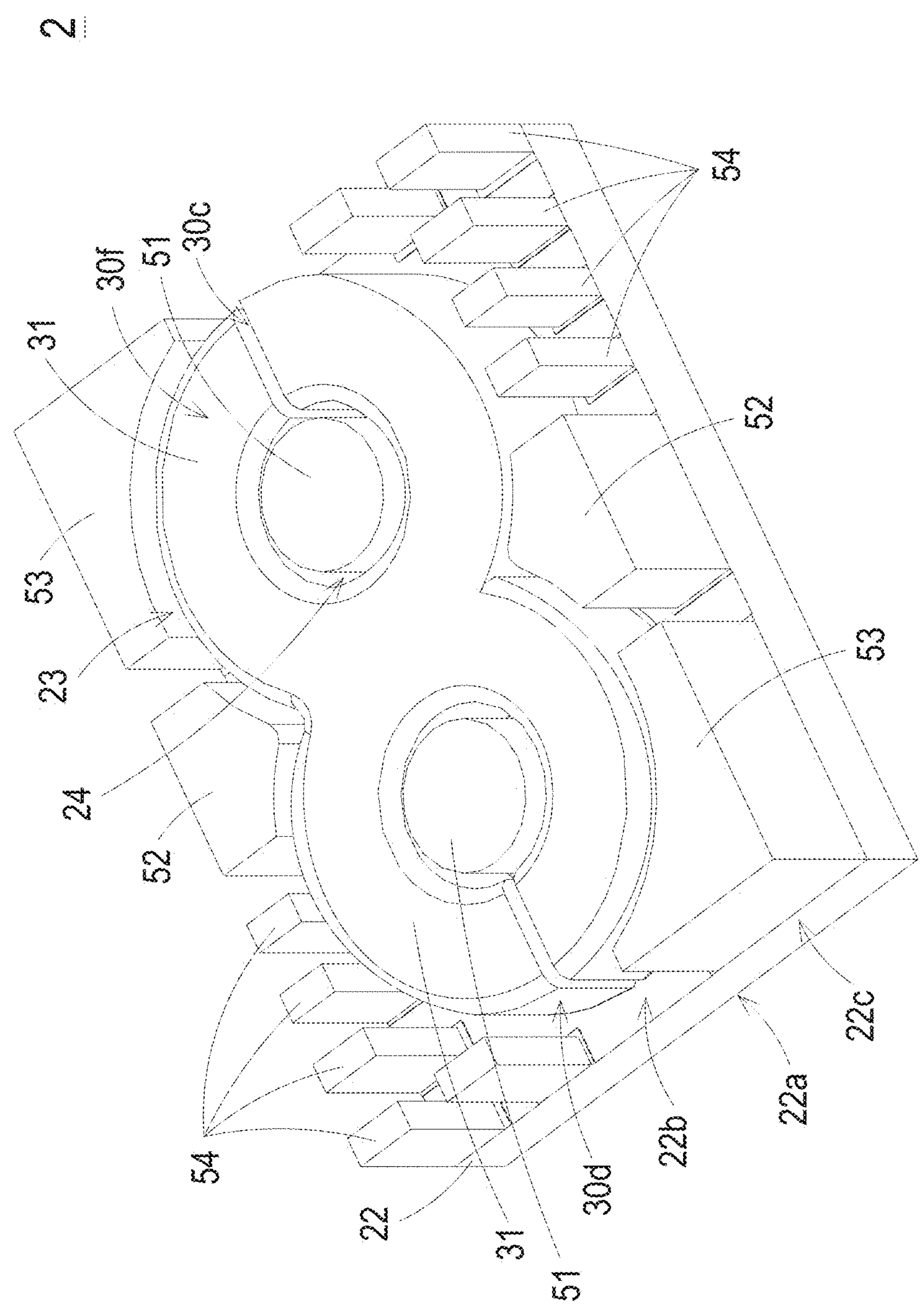


FIG. 3B

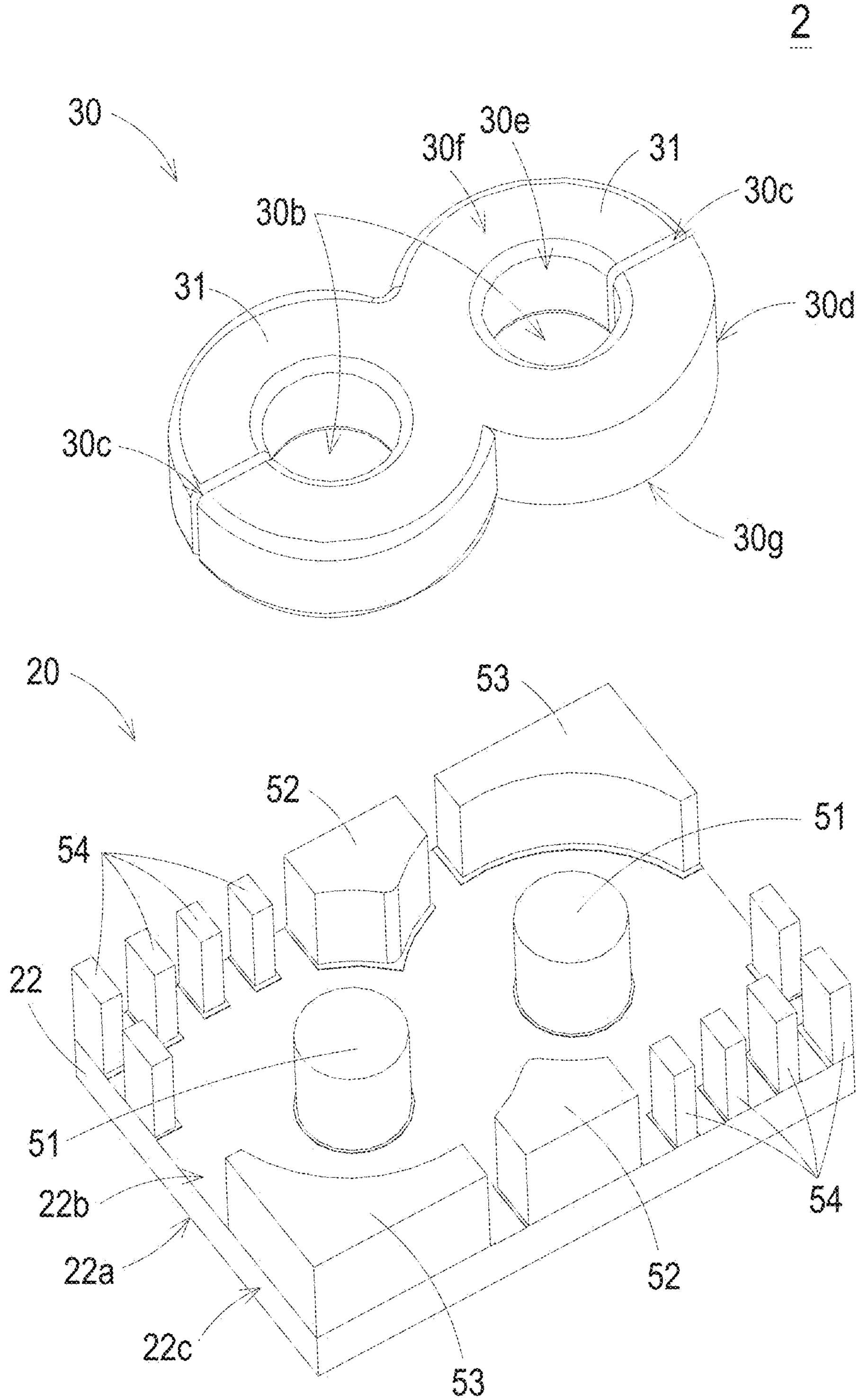

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FIG. 3C

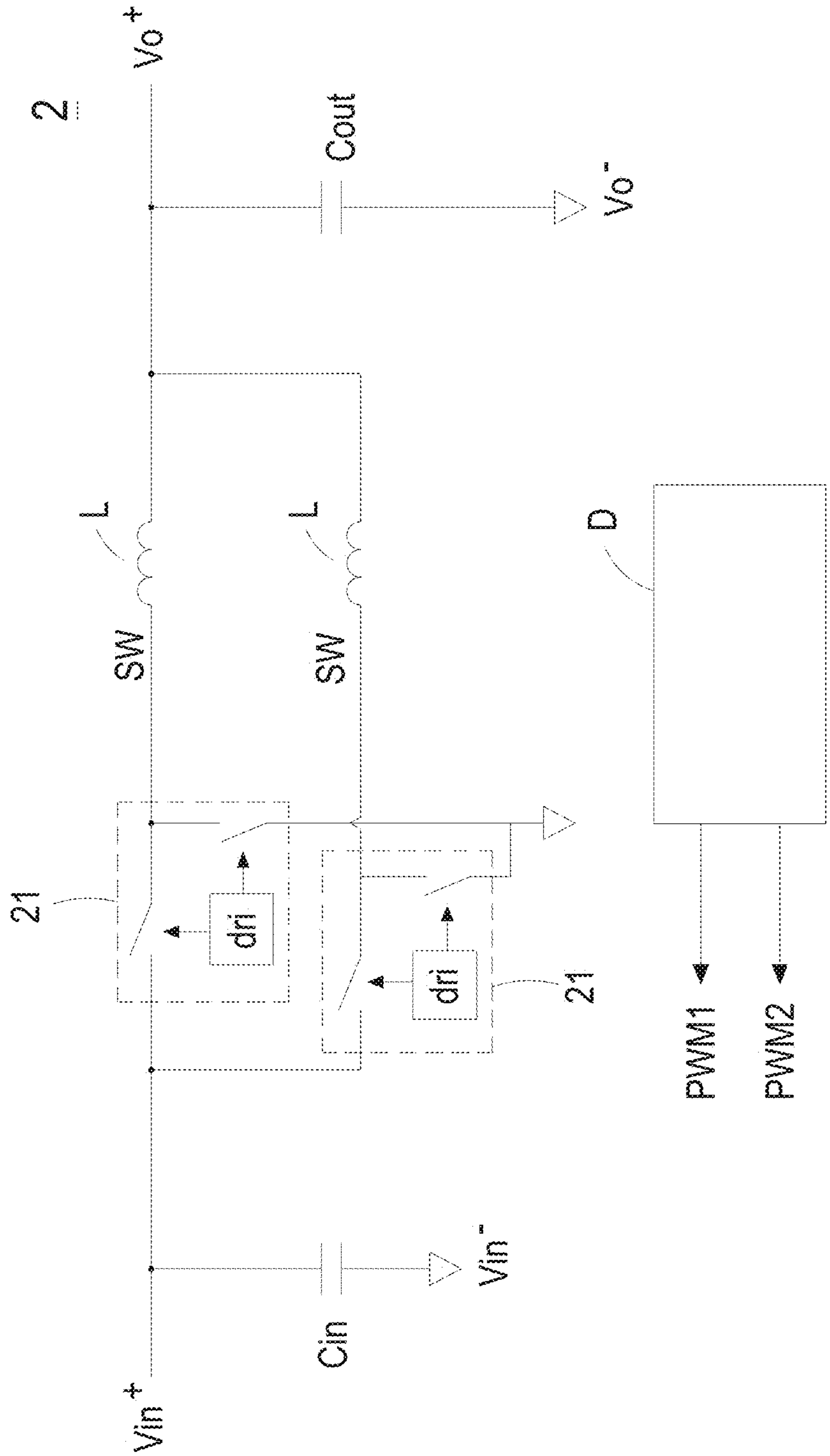

FIG. 4

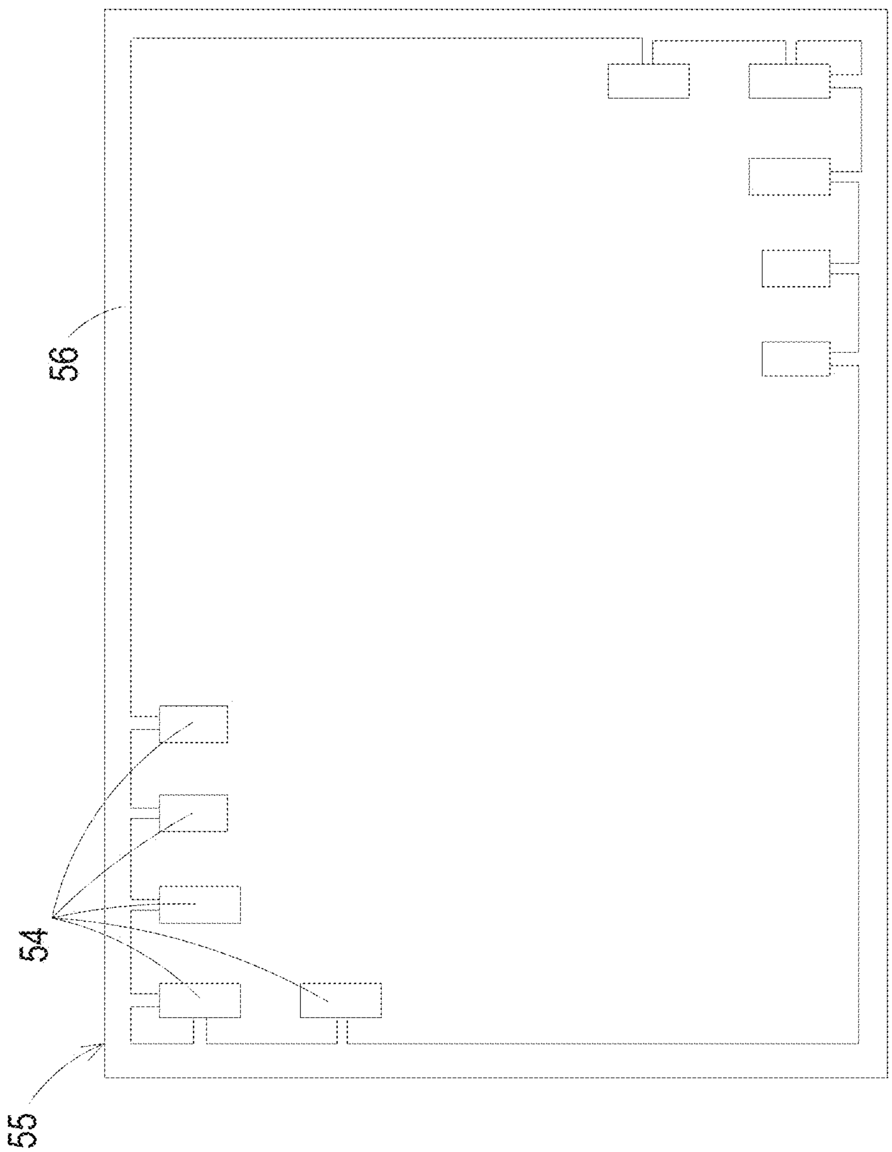

FIG. 5

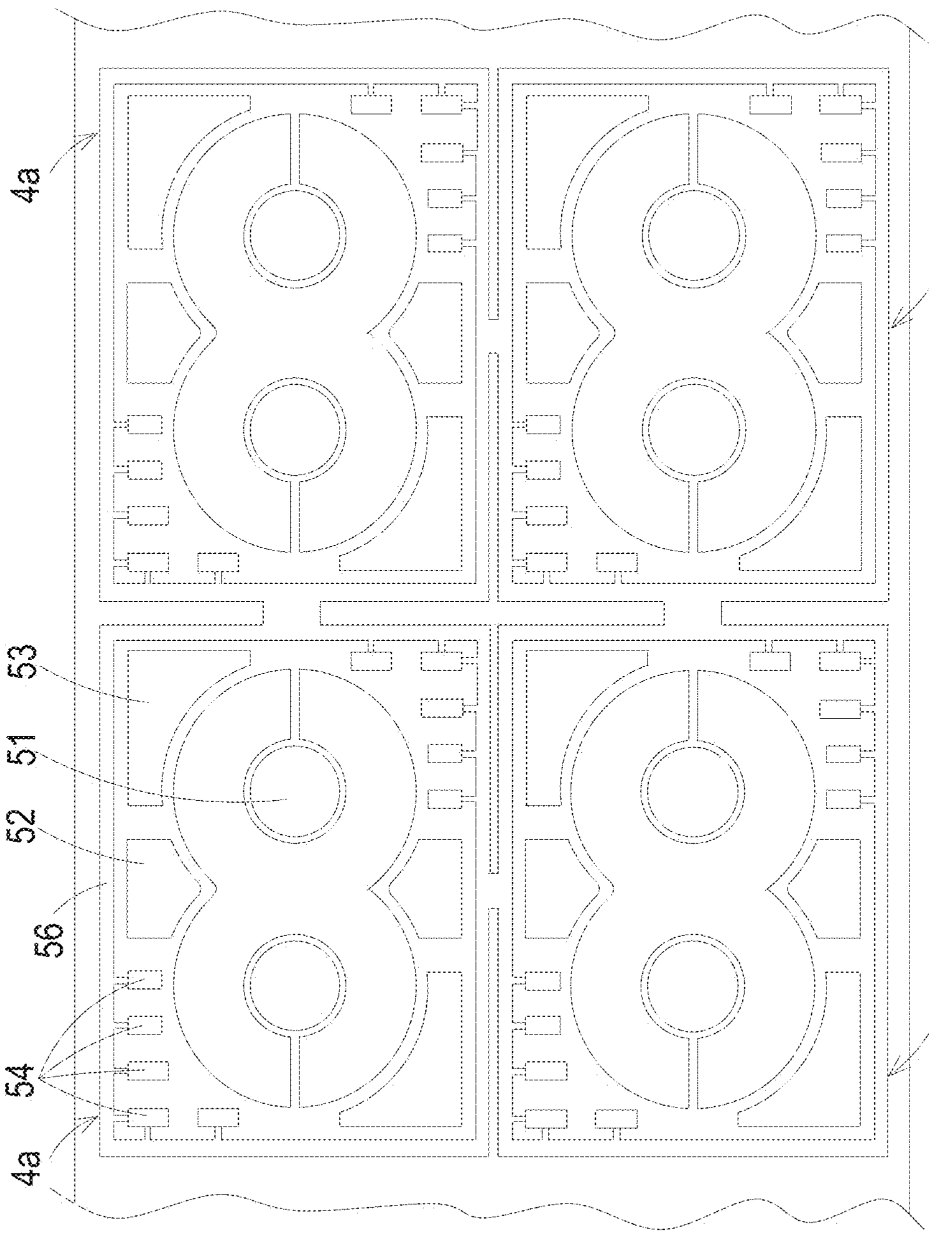

FIG. 6

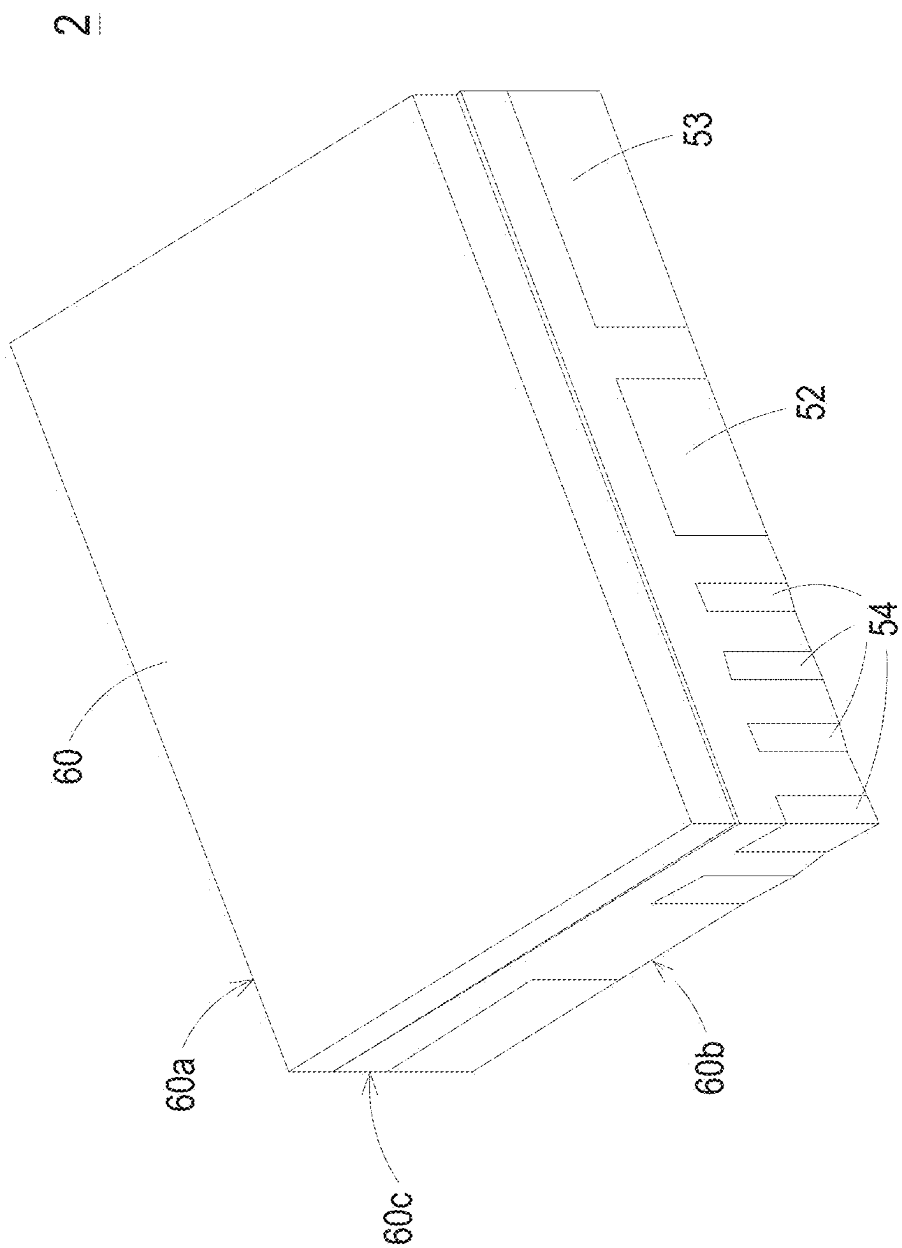

FIG. 7A

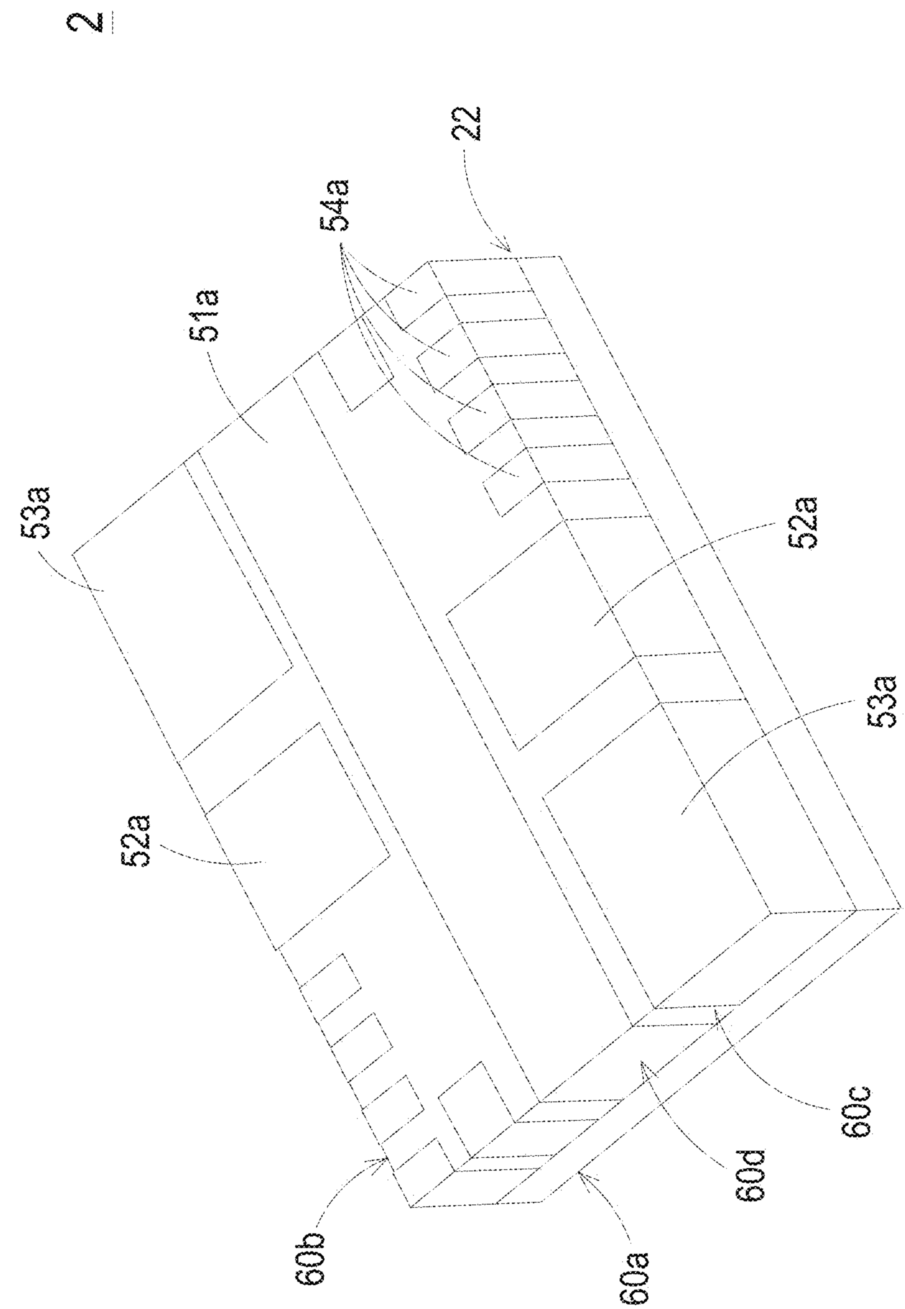

FIG. 7B

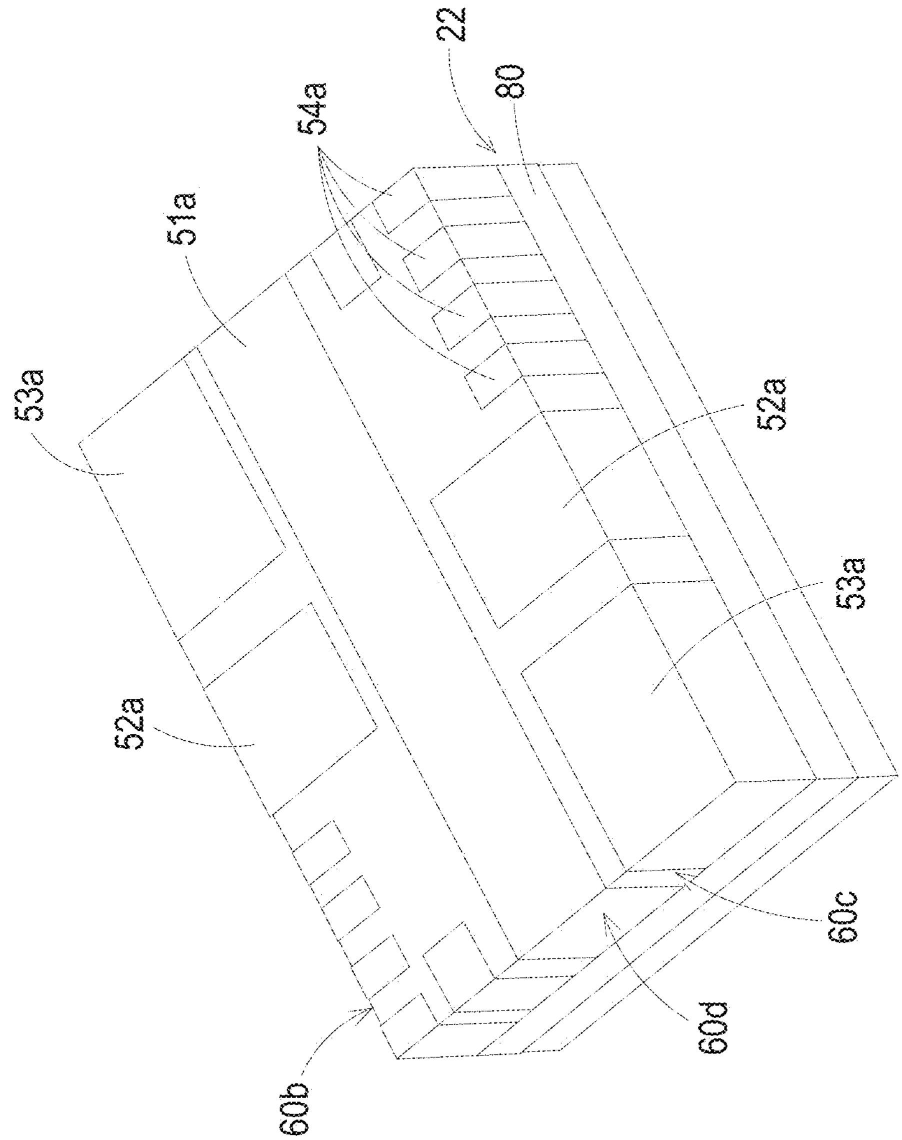

FIG. 8

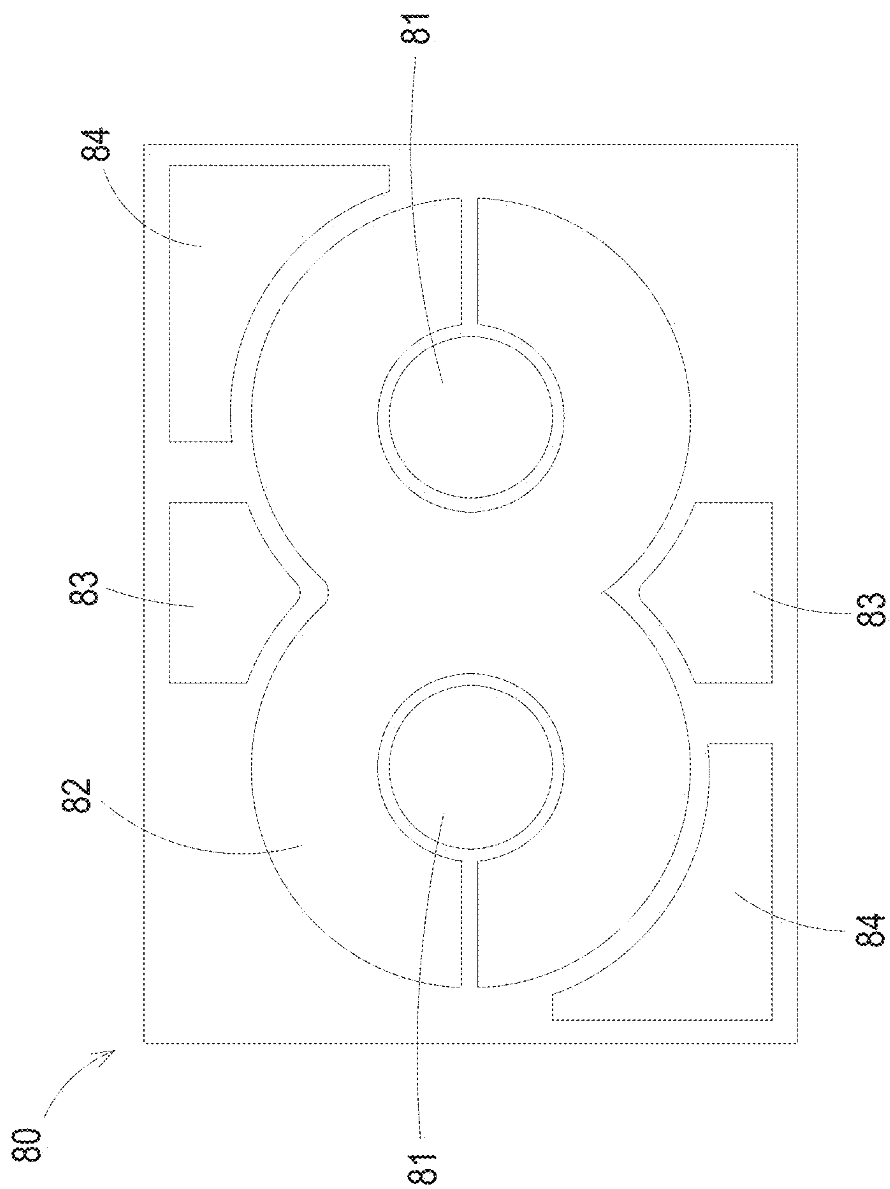

FIG. 9

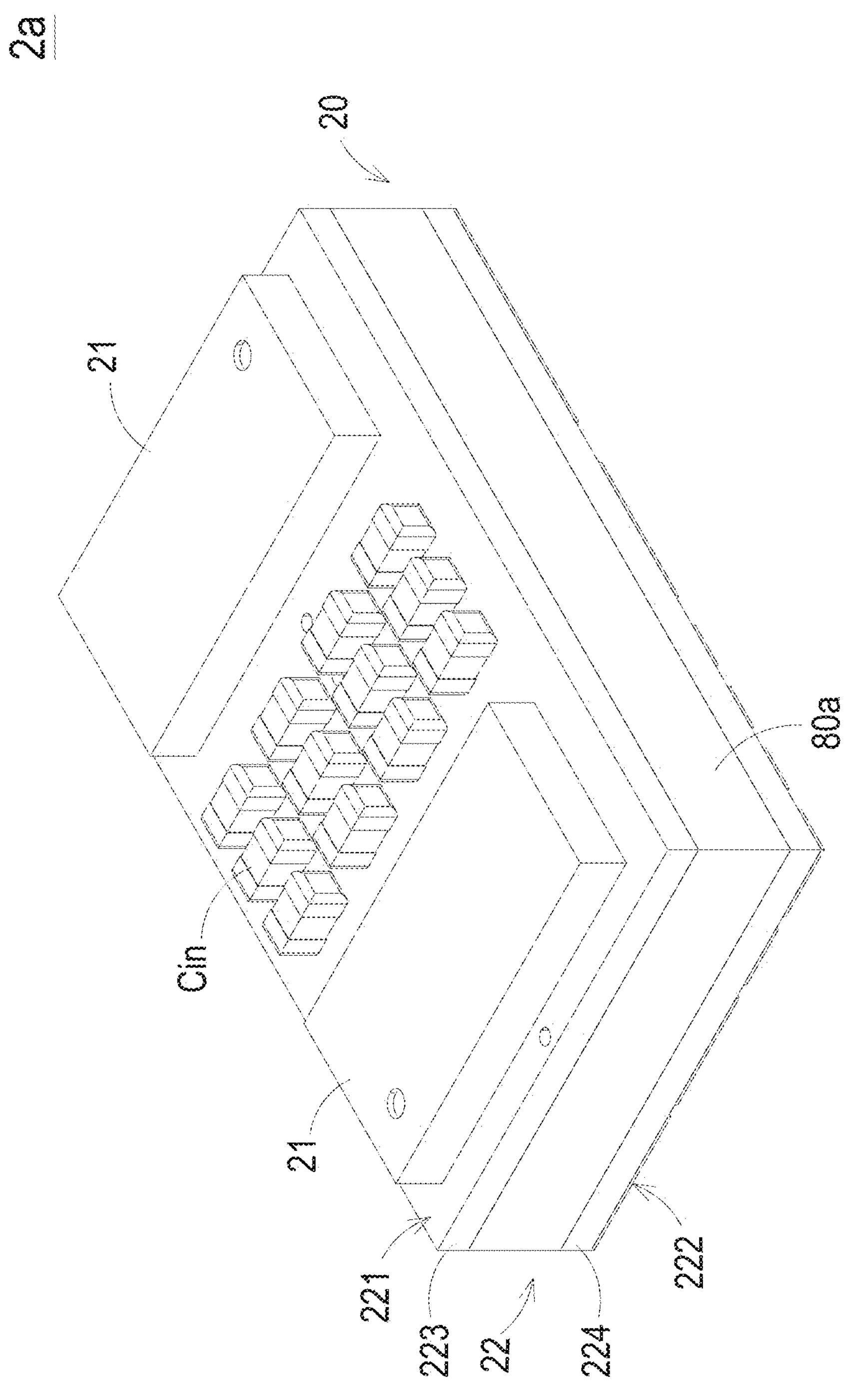

FIG. 10A

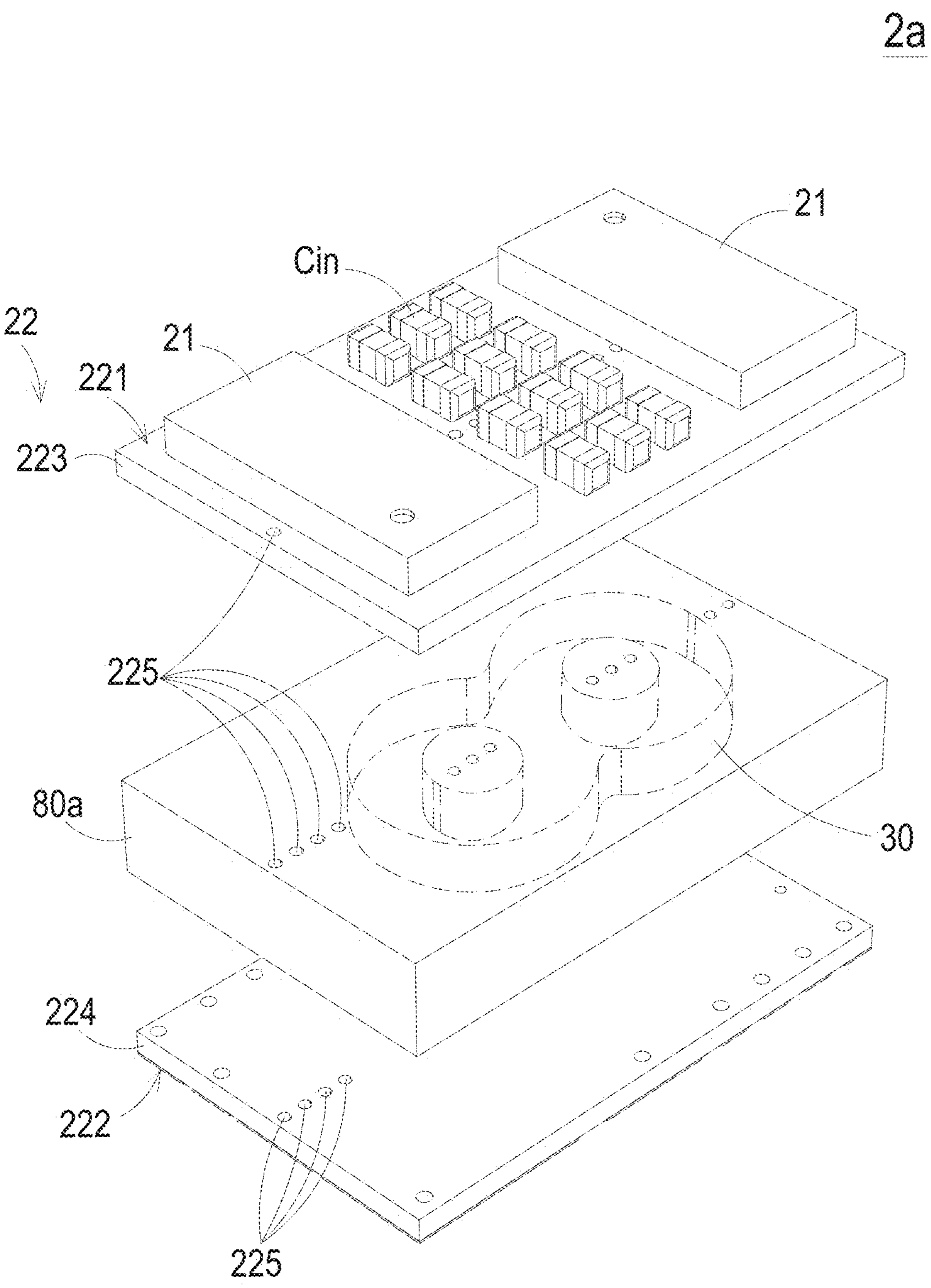

FIG. 10B

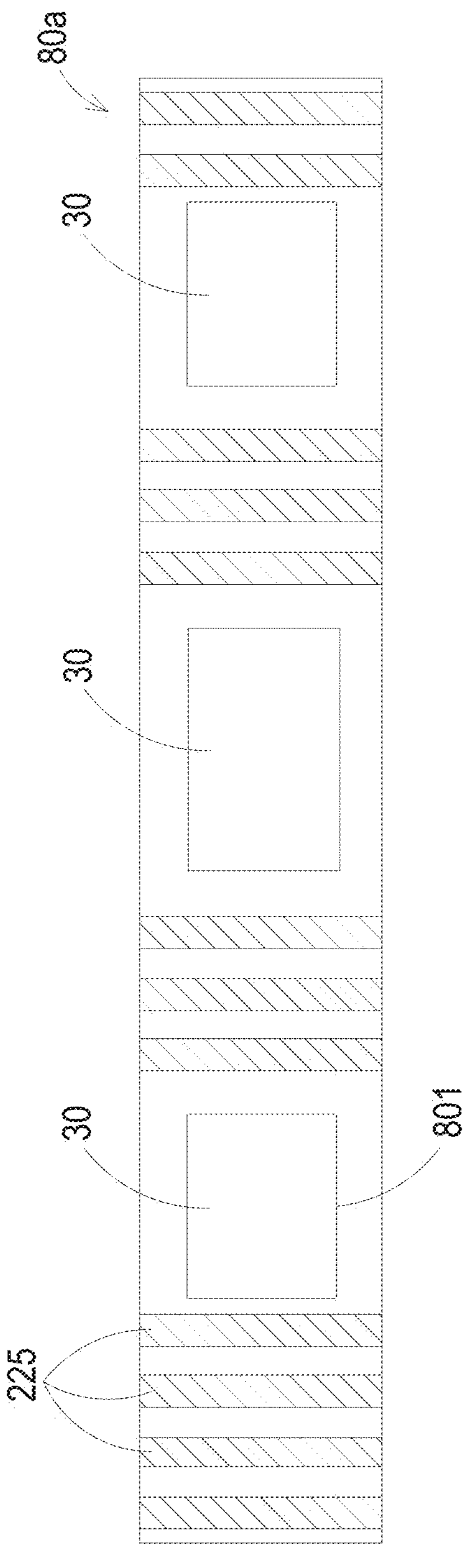

FIG. 10C

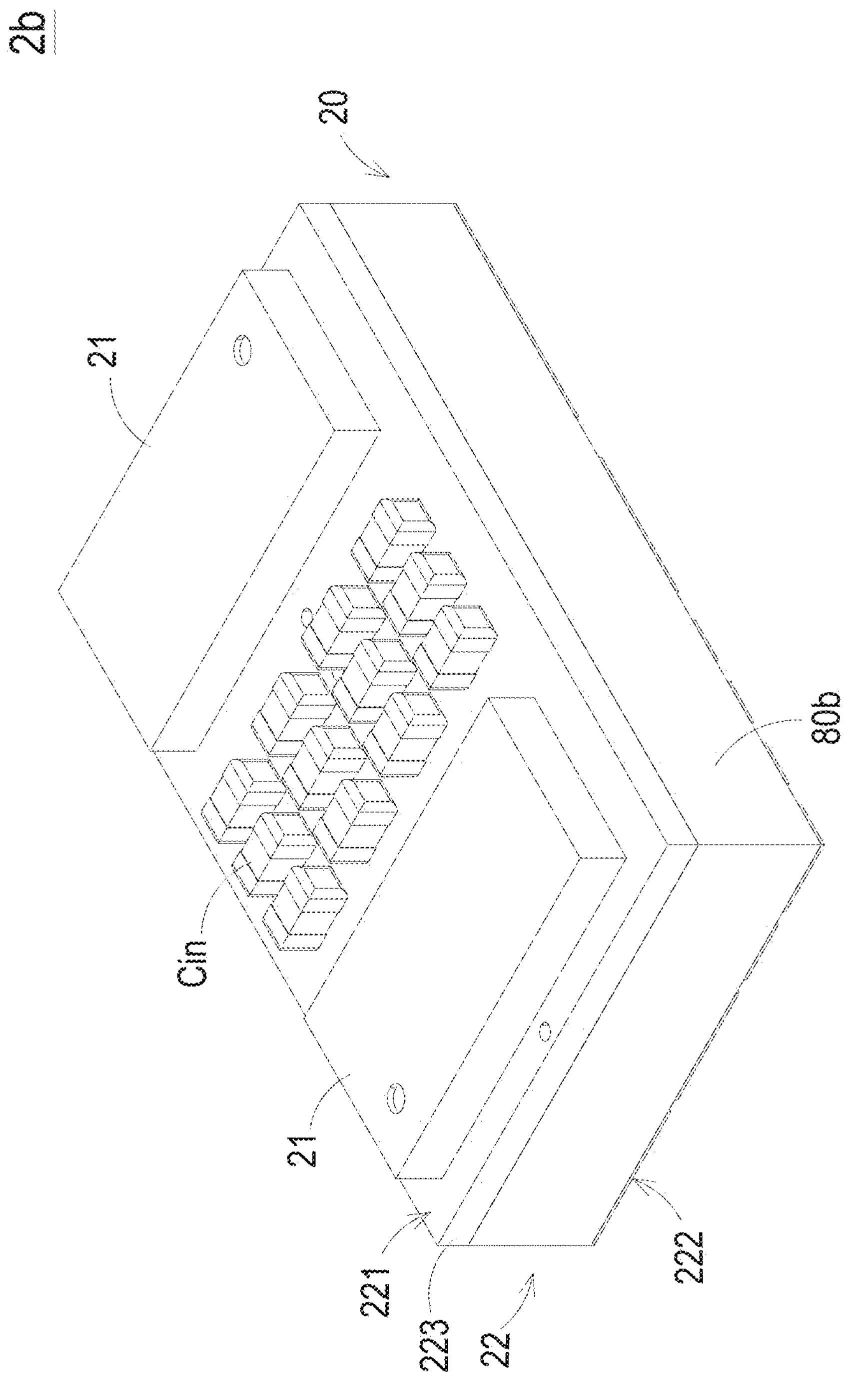

FIG. 11A

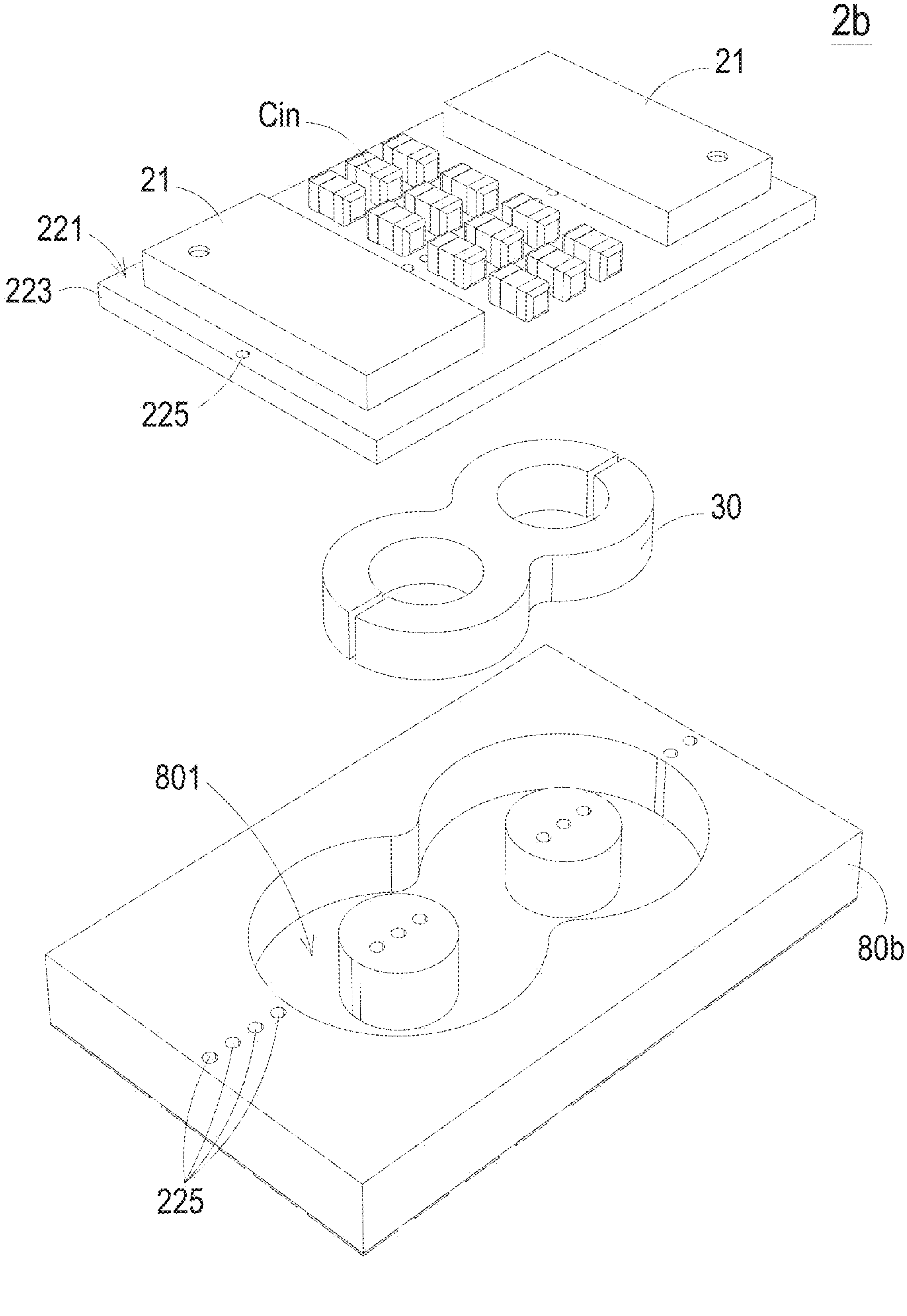

FIG. 11B

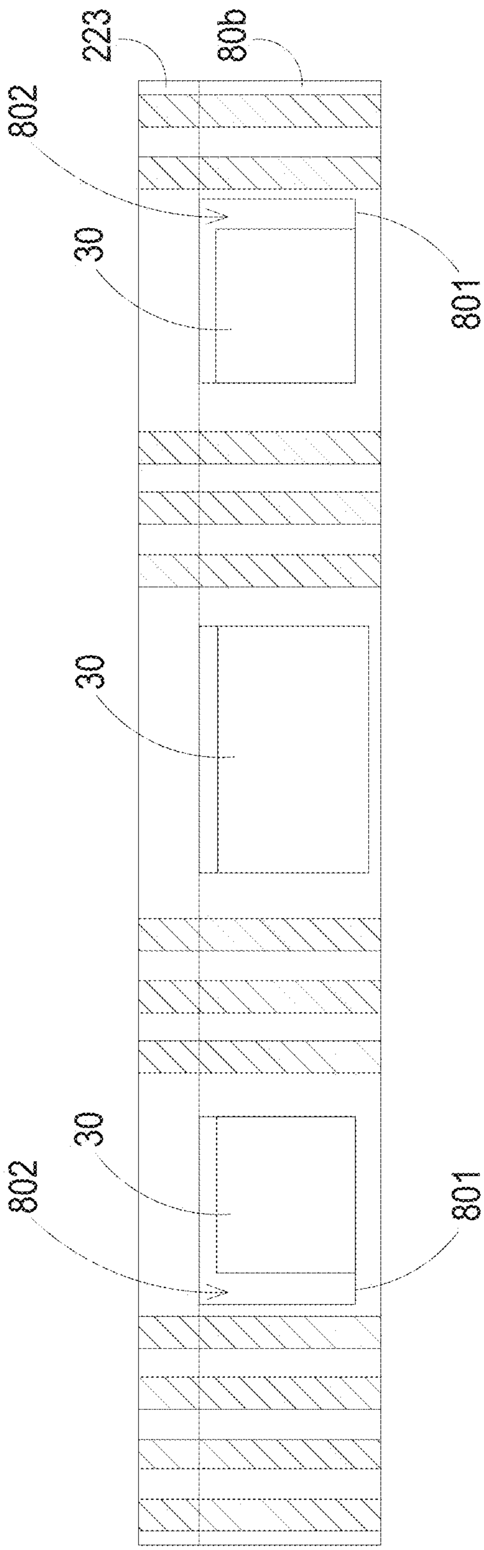

FIG. 11C

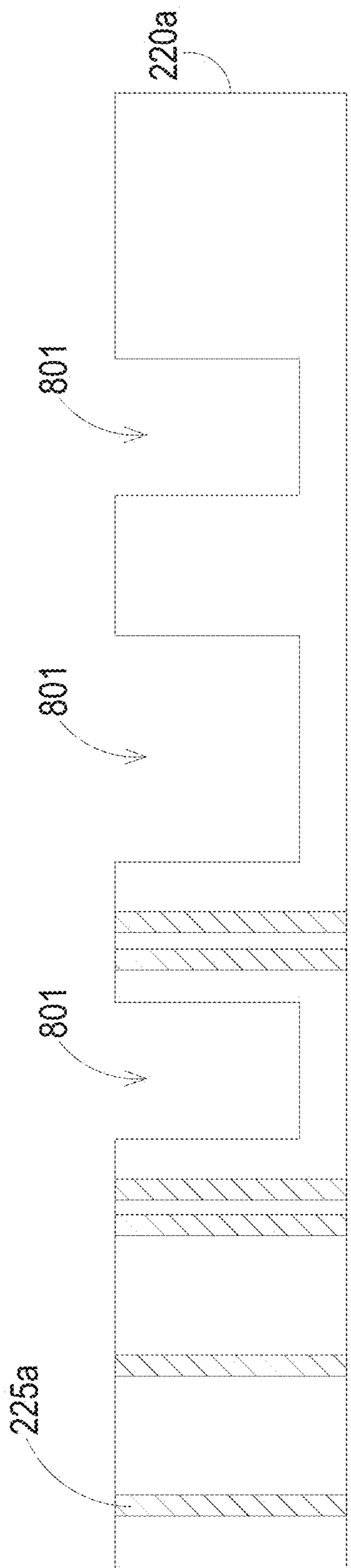

FIG. 11D

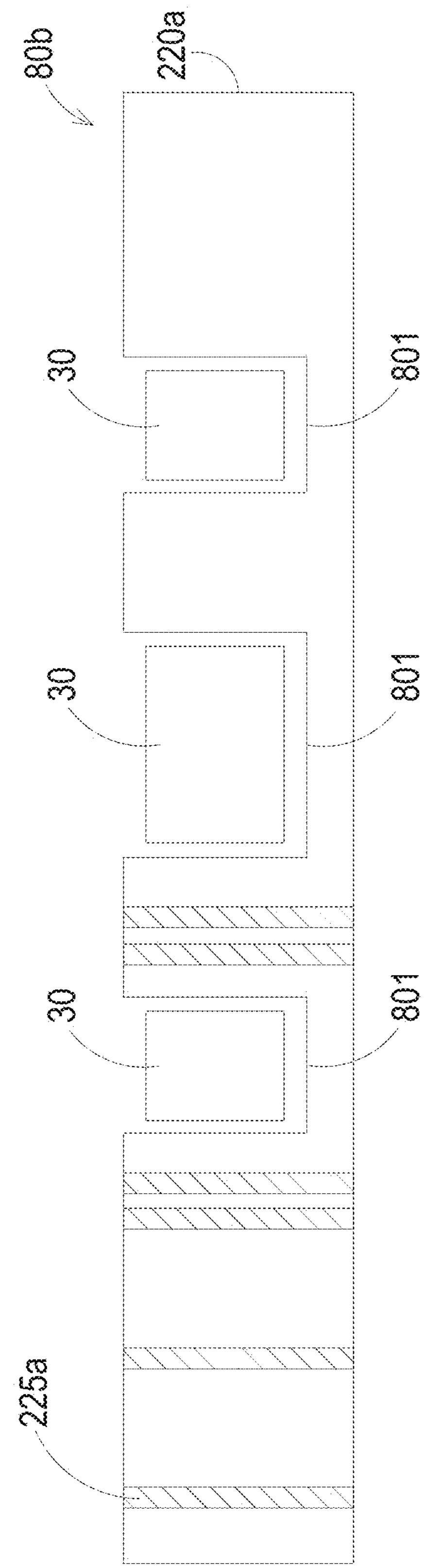

FIG. 11E

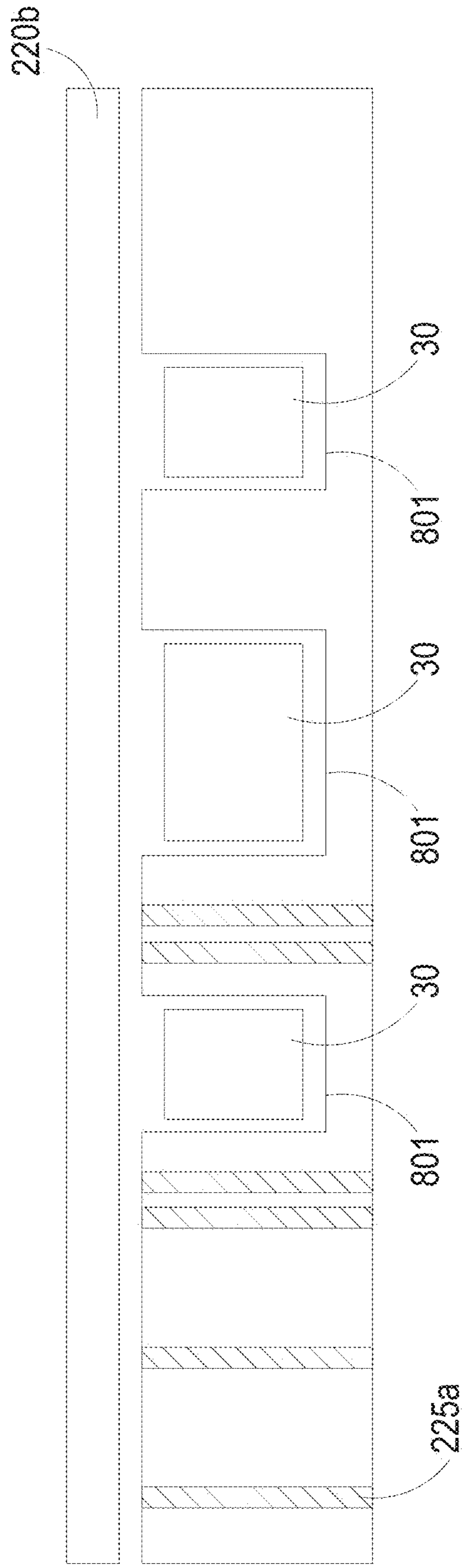

FIG. 11F

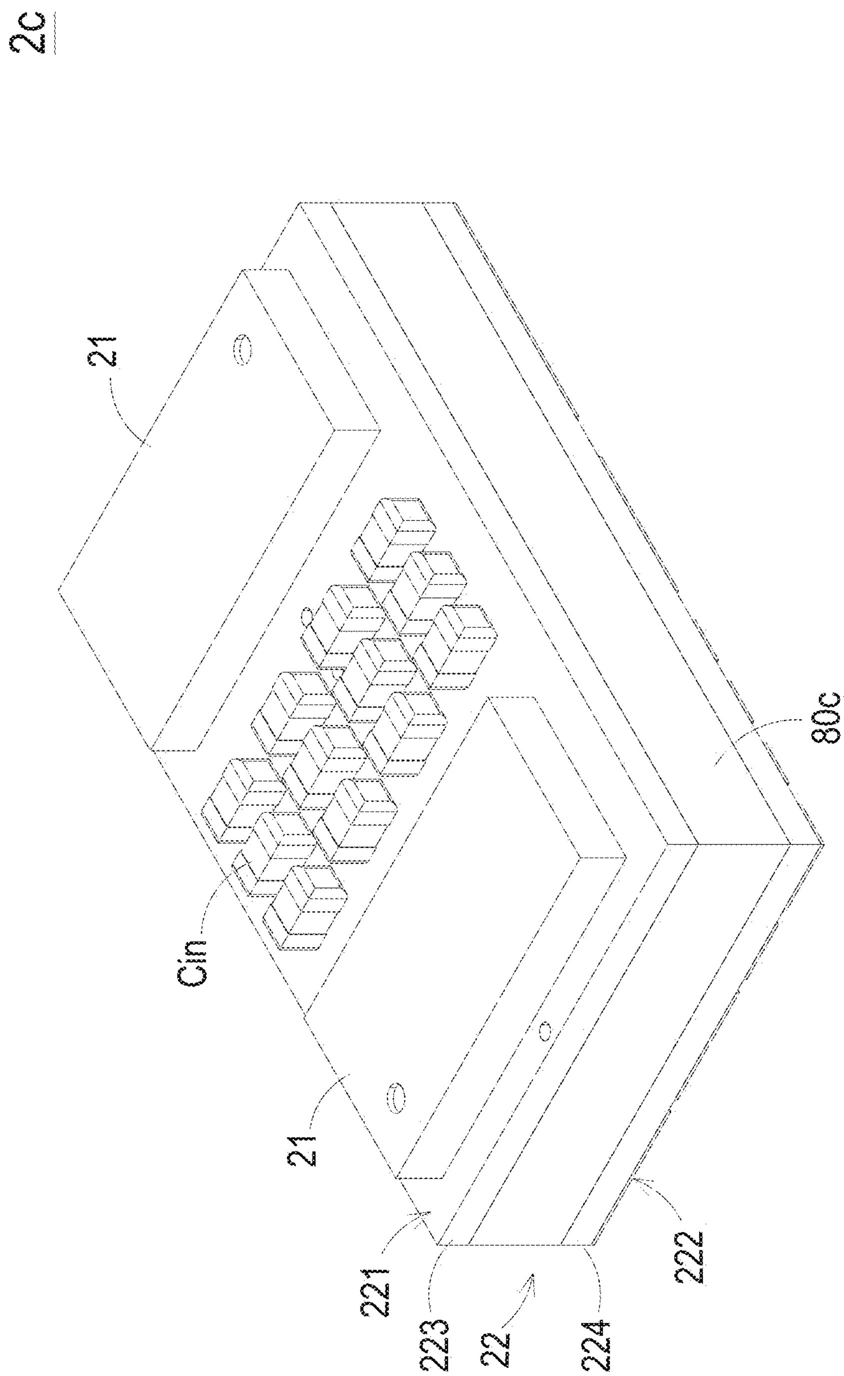

FIG. 12A

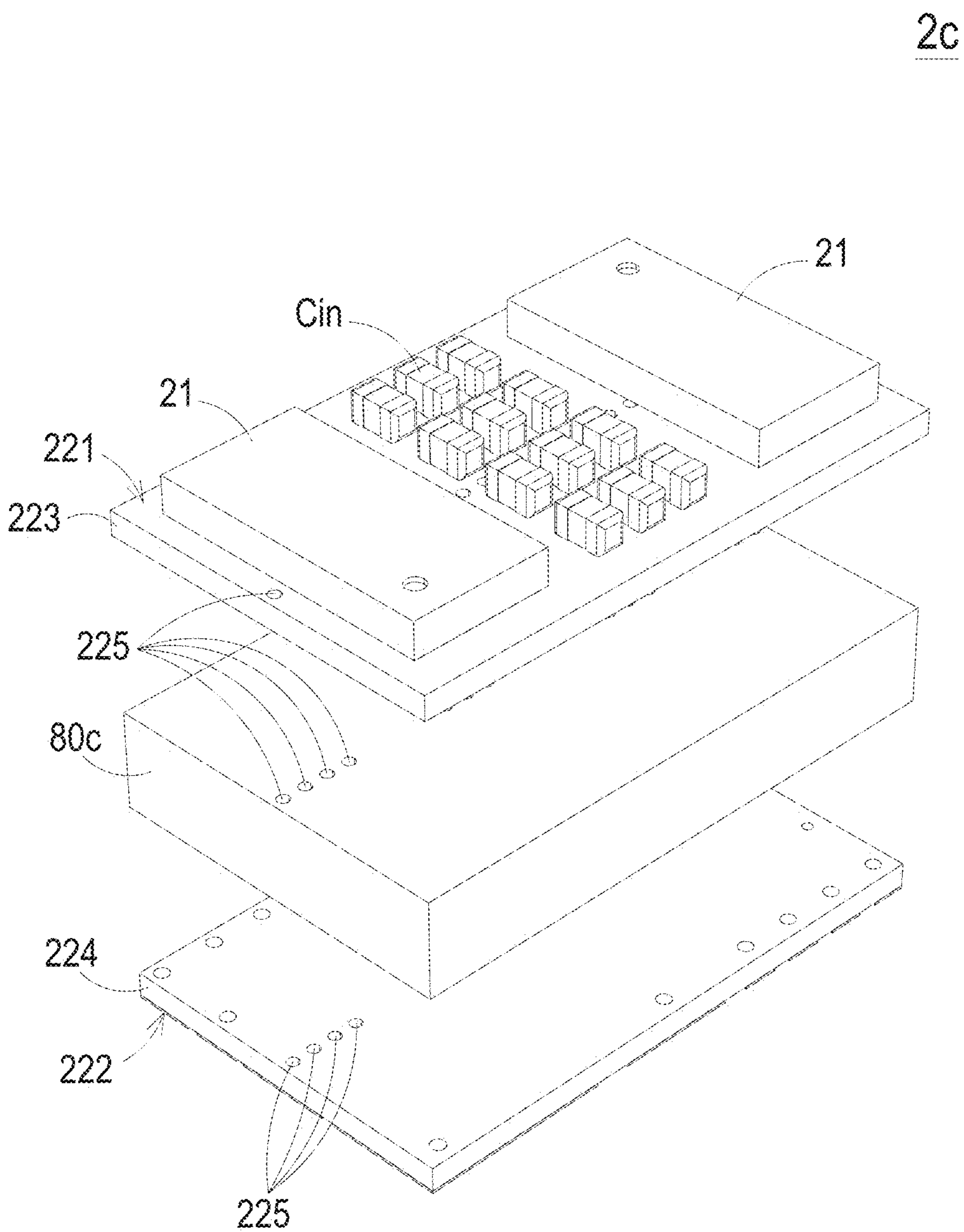

FIG. 12B

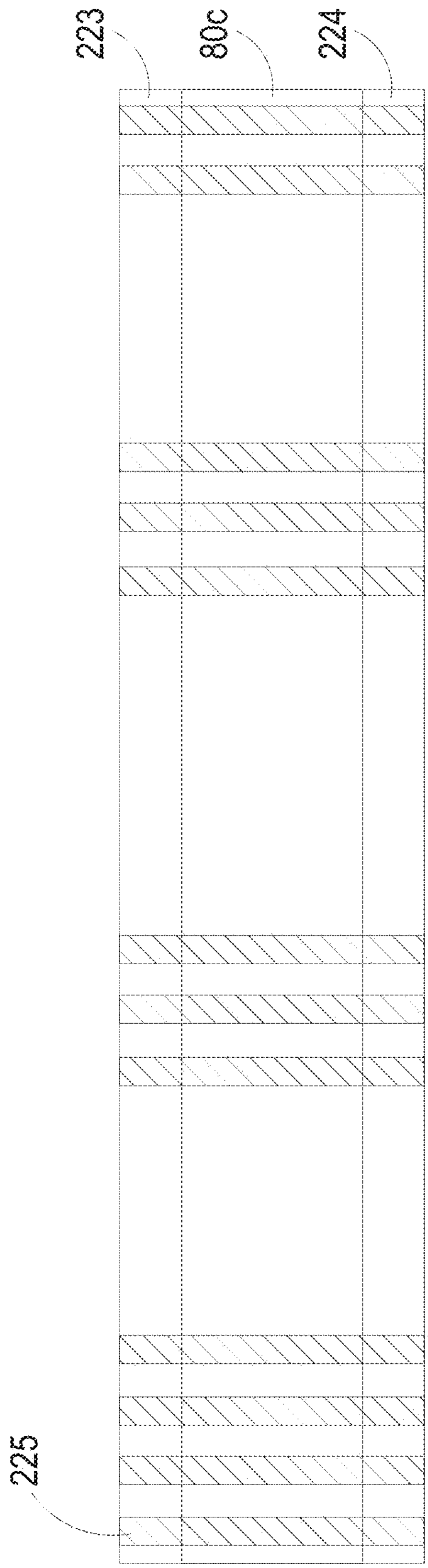

FIG. 12C

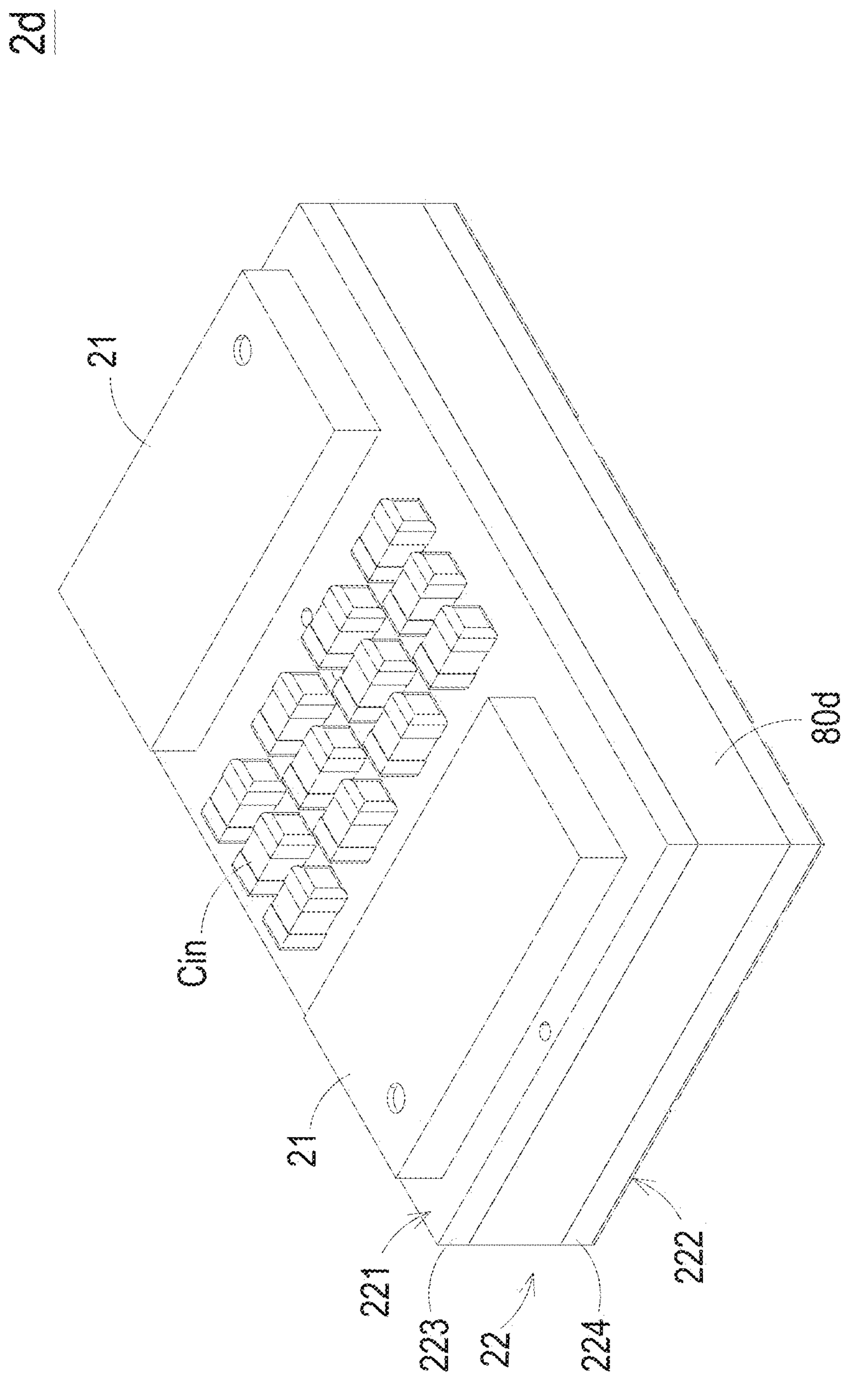

FIG. 13A

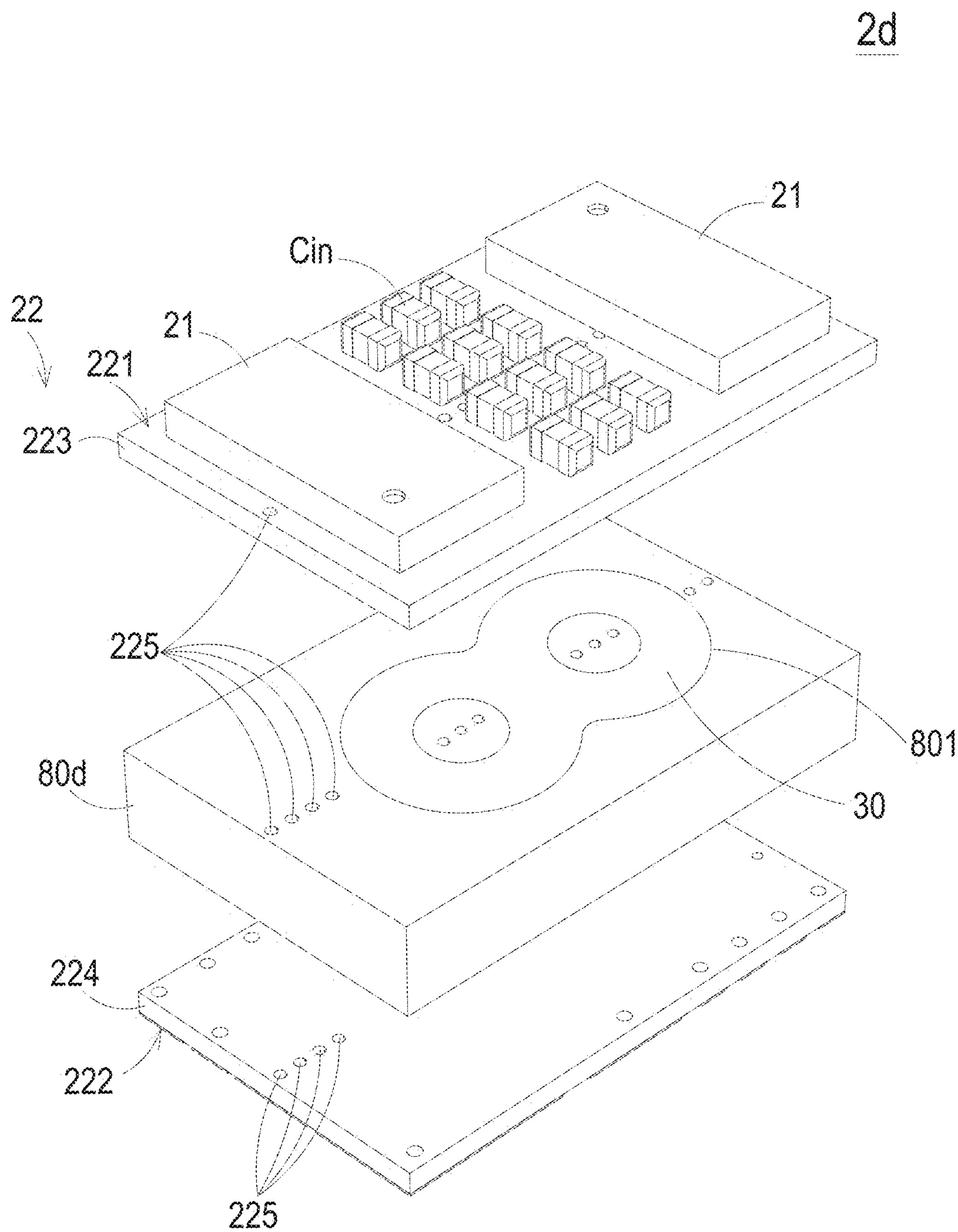

FIG. 13B

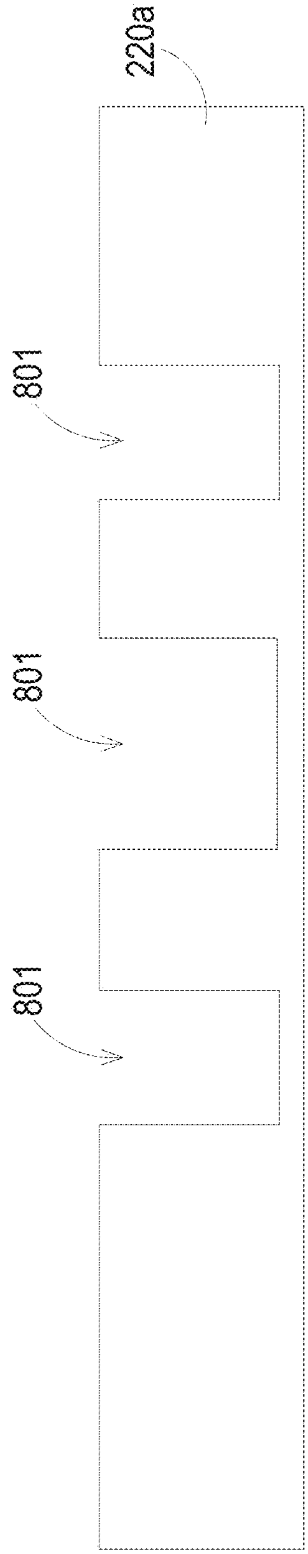

FIG. 13C

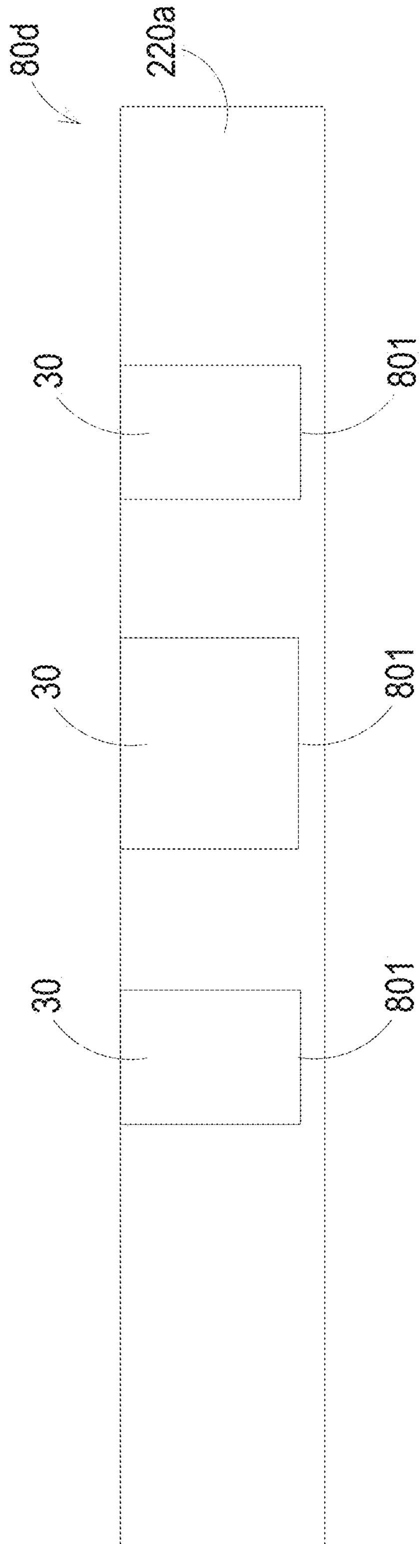

FIG. 13D

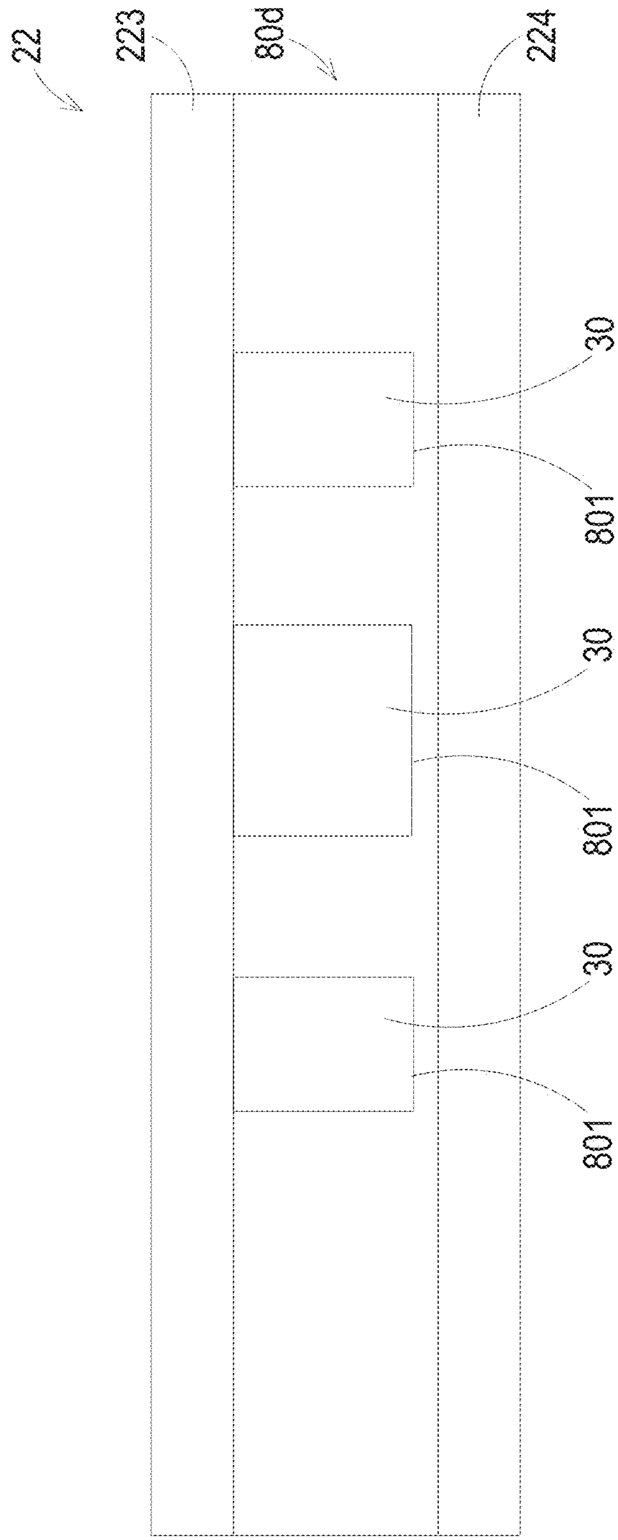

FIG. 13E

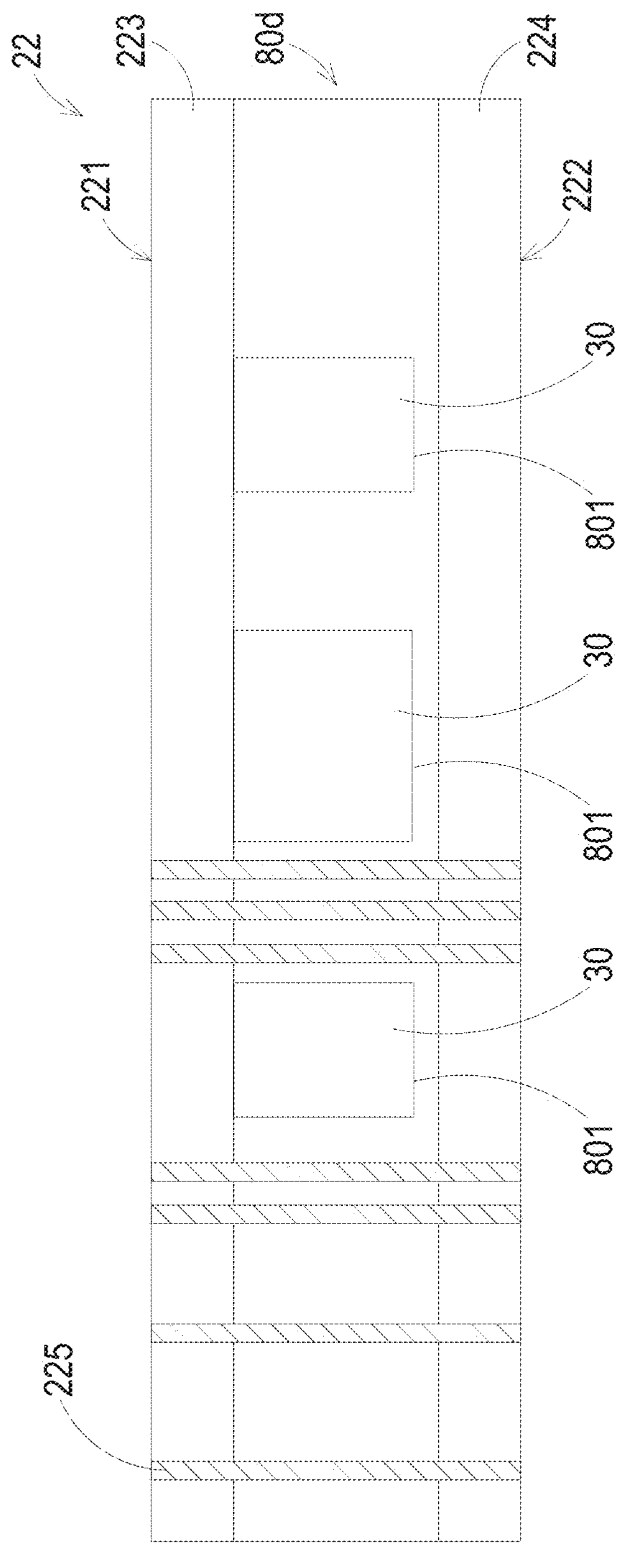

FIG. 13F

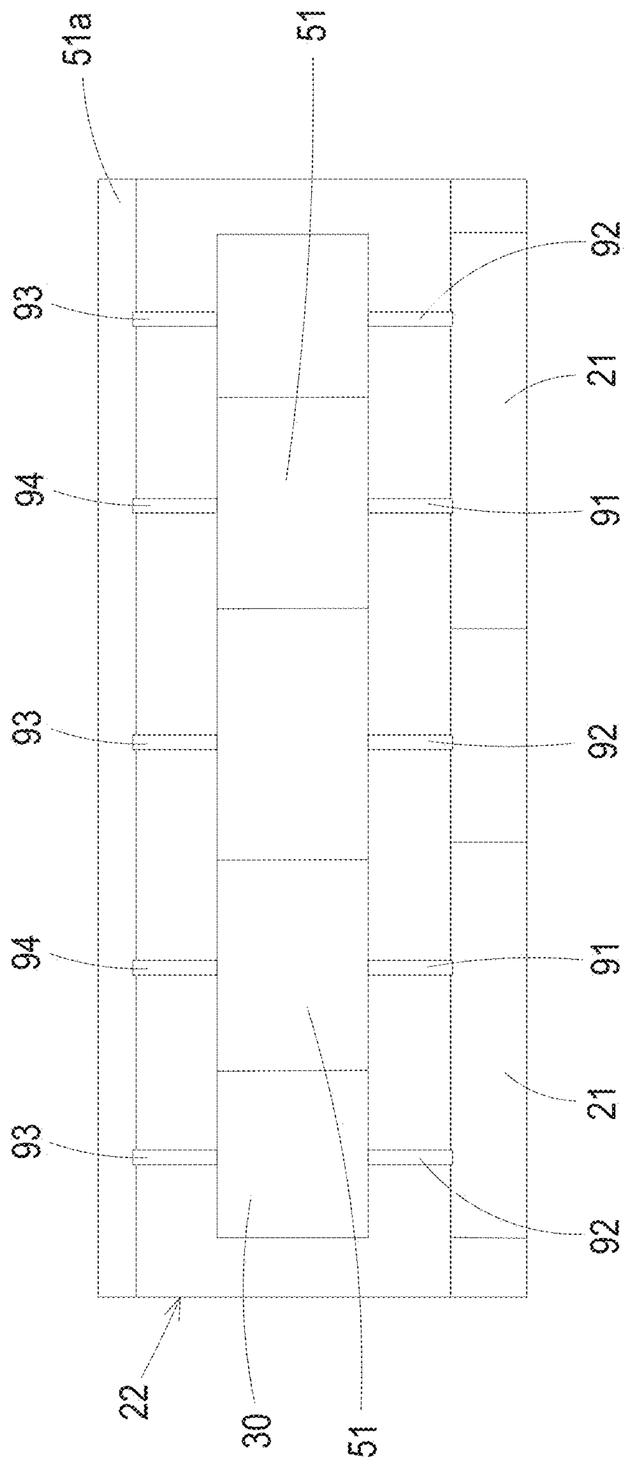

FIG. 14A

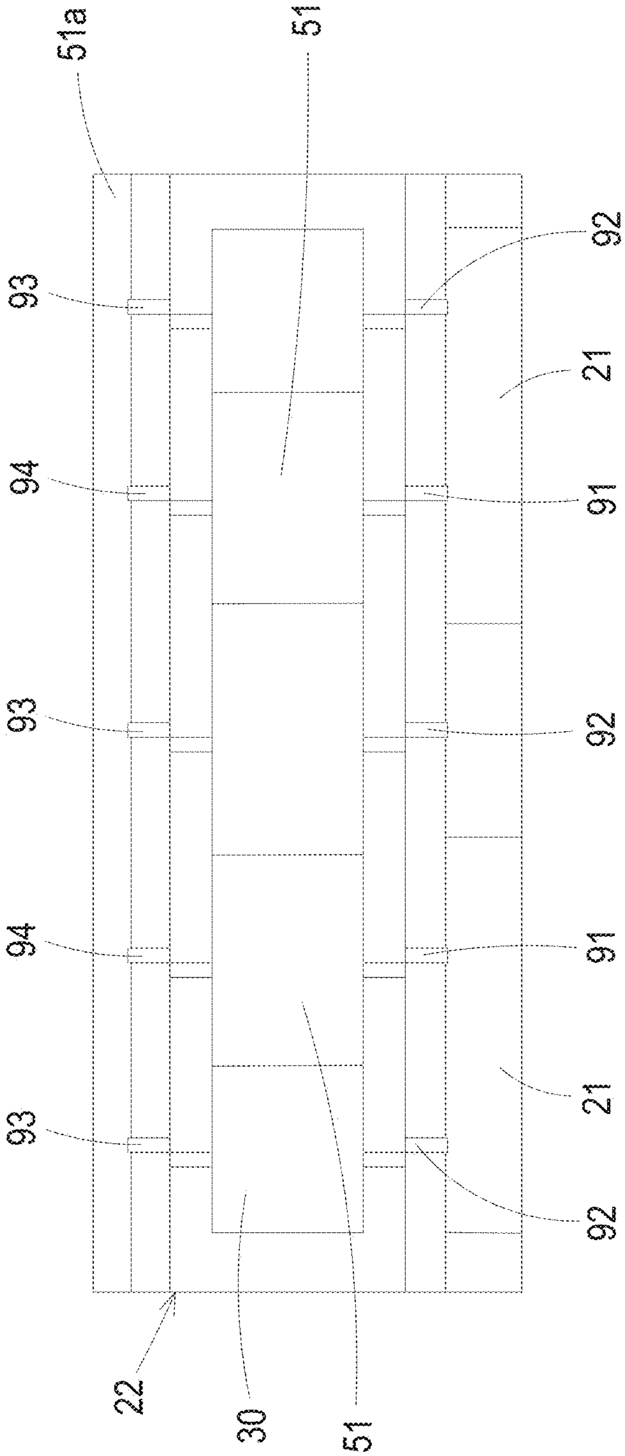

FIG. 14B

VOLTAGE REGULATOR MODULE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a Continuation-in-Part application of U.S. patent application Ser. No. 18/427,548 filed on Jan. 30, 2024, which is a Continuation application of U.S. patent application Ser. No. 17/530,790 filed on Nov. 19, 2021 and issued as U.S. Pat. No. 11,923,773 on Mar. 5, 2024, which is a Continuation application of U.S. patent application Ser. No. 16/810,406 filed on Mar. 5, 2020 and issued as U.S. Pat. No. 11,212,919 on Dec. 28, 2021, which claims priority to China patent application No. 201910205050.6 filed on Mar. 18, 2019. The entireties of the above-mentioned patent applications are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

[0002] The present disclosure relates to a voltage regulator module, and more particularly to a voltage regulator module with reduced volume.

BACKGROUND OF THE INVENTION

[0003] Please refer to FIGS. **1**A and **1**B. FIG. **1**A schematically illustrates the structure of a conventional electronic device. FIG. **1**B schematically illustrates the structure of a voltage regulator module of the electronic device as shown in FIG. **1**A. As shown in FIGS. **1**A and **1**B, the electronic device **1** has a horizontal layout structure. The electronic device **1** includes a central processing unit (CPU) **11**, a voltage regulator module **12**, a system board **13** and an output capacitor **14**. The voltage regulator module **12** is used for converting an input voltage into a regulated voltage and providing the regulated voltage to the central processing unit **11**. The system board **13** has a first surface and a second surface, which are opposed to each other. The voltage regulator module **12** and the central processing unit **11** are disposed on the first surface of the system board **13**. For meeting the load dynamic switching requirements, the output terminal of the voltage regulator module **12** is located near the input terminal of the central processing unit **11**. The output capacitor **14** is disposed on the second surface of the system board **13**. The output capacitor **14** is located beside the input terminal of the central processing unit **11**.

[0004] The voltage regulator module **12** further includes a printed circuit board **15** and a magnetic element **16**. The magnetic element **16** is disposed on the printed circuit board **15**. Moreover, a switch element is disposed in a vacant space between the printed circuit board **15** and the magnetic element **16**. The printed circuit board **15** is disposed on the first surface of the system board **13**. The heat from the voltage regulator module **12** can be transferred to the system board **13** through the printed circuit board **15**. Moreover, the heat is dissipated away through a heat dissipation mechanism (not shown) of the system board **13**.

[0005] Recently, the required current for the central processing unit **11** is gradually increased. In addition, the trend of the volume of the electronic device is toward miniaturization. Since the central processing unit **11** and the voltage regulator module **12** are located at the same side of the system board **13**, the electronic device cannot meet the load dynamic switching requirements.

[0006] For reducing the volume of the electronic device and effectively enhancing the dynamic switching performance of the voltage regulator module, another electronic device is disclosed. FIG. **2** schematically illustrates the structure of another conventional electronic device. The electronic device **1'** of FIG. **2** has the vertical layout structure. The voltage regulator module **12** is disposed on the second surface of the system board **13**, so that the voltage regulator module **12** and the central processing unit **11** are disposed on opposed surfaces of the system board **13**. Consequently, the volume of the electronic device **1'** is effectively reduced. Moreover, since the output capacitor (not shown) is located near the output terminal of the voltage regulator module **12** and the input terminal of the central processing unit **11**, the dynamic switching performance of the voltage regulator module **12** is enhanced.

[0007] Although the dynamic switching performance of the voltage regulator module of the electronic device **1'** as shown in FIG. **2** is enhanced, there are still some drawbacks. For example, since the magnetic element **16** and the switch element of the voltage regulator module **12** are disposed on the same side of the printed circuit board **15** and the switch element is disposed in the vacant space between the printed circuit board **15** and the magnetic element **16**, it is difficult to optimize the structure and the size of the magnetic element **16** of the voltage regulator module **12**. In other words, the size of the voltage regulator module **12** in the electronic device **1'** is still large. Moreover, when the voltage regulator module **12** on the system board **13** undergoes a reflow soldering process, the inner components to be reheated are possibly detached or shifted.

[0008] Therefore, there is a need of providing an improved voltage regulator module in order to overcome the drawbacks of the conventional technologies.

SUMMARY OF THE INVENTION

[0009] An object of the present disclosure provides a voltage regulator module with reduced size.

[0010] Another object of the present disclosure provides a voltage regulator module that is manufactured by a simplified fabricating process.

[0011] In accordance with an aspect of the present disclosure, a voltage regulator module is provided. The voltage regulator module includes a circuit board assembly and a magnetic core assembly. The circuit board assembly includes a printed circuit board and at least one switch element. The printed circuit board comprises an upper surface, a lower surface, an inner layer and a plurality of conductive portion. The upper surface and the lower surface are opposite to each other. The inner layer is embedded within the printed circuit board. The plurality of conductive portions are electrically connected between the upper surface and the lower surface of the printed circuit board. The at least one switch element is disposed on the upper surface of the printed circuit board. The magnetic core assembly is disposed in the inner layer.

[0012] The above contents of the present disclosure will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. **1**A schematically illustrates the structure of a conventional electronic device;

[0014] FIG. **1**B schematically illustrates the structure of a voltage regulator module of the electronic device as shown in FIG. **1**A;

[0015] FIG. **2** schematically illustrates the structure of another conventional electronic device;

[0016] FIG. **3**A is a schematic perspective view illustrating a voltage regulator module according to a first embodiment of the present disclosure and taken along a viewpoint;

[0017] FIG. **3**B is a schematic perspective view illustrating the voltage regulator module of FIG. **3**A and taken along another viewpoint;

[0018] FIG. **3**C is a schematic exploded view illustrating the voltage regulator module of FIG. **3**B;

[0019] FIG. **4** is a schematic equivalent circuit diagram illustrating the voltage regulator module of FIG. **3**A;

[0020] FIG. **5** schematically illustrates a lead frame with the fourth copper posts;

[0021] FIG. **6** schematically illustrates the concept of using the lead frame as shown in FIG. **5** to produce a plurality of voltage regulator modules;

[0022] FIG. **7**A is a schematic perspective view illustrating a package structure of the voltage regulator module according to a first embodiment of the present disclosure;

[0023] FIG. **7**B is a schematic perspective view illustrating the package structure of the voltage regulator module as shown in FIG. **7**A and taken along another viewpoint;

[0024] FIG. **8** is a schematic perspective view illustrating a package structure of a voltage regulator module according to a second embodiment of the present disclosure;

[0025] FIG. **9** is a schematic top cross-sectional view illustrating the inner layer of the printed circuit board of the voltage regulator module as shown in FIG. **8**;

[0026] FIG. **10**A is a schematic perspective view illustrating a voltage regulator module according to a third embodiment of the present disclosure;

[0027] FIG. **10**B is a schematic exploded view illustrating the voltage regulator module of FIG. **10**A;

[0028] FIG. **10**C is a schematic cross-sectional view illustrating the inner layer of the voltage regulator module of FIG. **10**A;

[0029] FIG. **11**A is a schematic perspective view illustrating a voltage regulator module according to a fourth embodiment of the present disclosure;

[0030] FIG. **11**B is a schematic exploded view illustrating the voltage regulator module of FIG. **11**A;

[0031] FIG. **11**C is a schematic cross-sectional view illustrating the printed circuit board of the voltage regulator module of FIG. **11**A;

[0032] FIGS. **11**D to **11**F are schematic cross-sectional views illustrating a process of manufacturing the printed circuit board of FIGS. **11**A to **11**C;

[0033] FIG. **12**A is a schematic perspective view illustrating a voltage regulator module according to a fifth embodiment of the present disclosure;

[0034] FIG. **12**B is a schematic exploded view illustrating the voltage regulator module of FIG. **12**A;

[0035] FIG. **12**C is a schematic cross-sectional view illustrating the printed circuit board of the voltage regulator module of FIG. **12**A;

[0036] FIG. **13**A is a schematic perspective view illustrating a voltage regulator module according to a sixth embodiment of the present disclosure;

[0037] FIG. **13**B is a schematic exploded view illustrating the voltage regulator module of FIG. **13**A;

[0038] FIGS. **13**C to **13**F are schematic cross-sectional views illustrating a process of manufacturing the printed circuit board of FIGS. **13**A to **13**B;

[0039] FIG. **14**A is a schematic cross-sectional view illustrating the package structure of the voltage regulator module as shown in FIG. **8**; and

[0040] FIG. **14**B is a schematic cross-sectional view illustrating a variant example of the package structure of the voltage regulator module as shown in FIG. **14**A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0041] The present disclosure will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed.

[0042] Please refer to FIGS. **3**A, **3**B, **3**C and **4**. FIG. **3**A is a schematic perspective view illustrating a voltage regulator module according to a first embodiment of the present disclosure and taken along a viewpoint. FIG. **3**B is a schematic perspective view illustrating the voltage regulator module of FIG. **3**A and taken along another viewpoint. FIG. **3**C is a schematic exploded view illustrating the voltage regulator module of FIG. **3**B. FIG. **4** is a schematic equivalent circuit diagram illustrating the voltage regulator module of FIG. **3**A. The voltage regulator module **2** may be disposed on an electronic device and electrically connected with a system board (not shown) within the electronic device through a welding means. The voltage regulator module **2** is at least one phase buck converter and includes at least one switch elements **21**, at least one inductor L, at least one input capacitor Cin and at least one output capacitor Cout. For example, the switch element **21** is a driver and metal-oxide-semiconductor field-effect transistor (also referred as a Dr.MOS element **21**). In case that the central processing unit of the electronic device requires a large amount of current, the voltage regulator module **2** is a multi-phase buck converter. Consequently, the capability of the voltage regulator module **2** to output electricity is effectively enhanced.

[0043] In the embodiment of FIG. **4**, the voltage regulator module **2** is a two-phase buck converter. The voltage regulator module **2** includes two Dr.MOS elements **21** and two inductors L. Each Dr.MOS element **21** and a first terminal SW of the corresponding inductor L are electrically connected with each other in series to define a phase buck circuit. In this embodiment, the voltage regulator module **2** includes two phase buck circuits. The input side of the voltage regulator module **2** includes a positive input terminal Vin+ and a negative input terminal Vin−. The first terminals of the two phase buck circuits are connected with each other in parallel and electrically connected with the input capacitor Cin. The output side of the voltage regulator module **2** includes a positive output terminal Vo+ and a negative output terminal Vo−. The negative input terminal Vin− and the negative output terminal Vo− are electrically connected with a common terminal. The second terminals of the two phase buck circuits are connected with each other in parallel and electrically connected with the output capacitor Cout. A first terminal of the output capacitor Cout is electrically connected with the positive output terminal Vo+ of the voltage regulator module **2**. A second terminal of the output capacitor Cout is electrically connected with the negative output terminal Vo− of the voltage regulator module **2**. A first terminal of the input capacitor Cin is electrically connected with the positive input terminal Vin+ of the voltage regulator module **2**. A second terminal of the input capacitor Cin is electrically connected with the negative input terminal Vin− of the voltage regulator module **2**.

[0044] In an embodiment, each Dr.MOS element **21** includes a switch and a driver for driving the switch. The voltage regulator module **2** further includes a control circuit D. After the control circuit D samples the output voltage of the voltage regulator module **2** and the output current of each phase buck circuit, the control circuit D generates two pulse width modulation signals PWM**1** and PWM**2**. The phase difference between the two pulse width modulation signals PWM**1** and PWM**2** is 180 degree. The first phase buck circuit and the second phase buck circuit are controlled according to the first pulse width modulation signal PWM**1** and the second pulse width modulation signal PWM**2**, respectively.

[0045] Please refer to FIG. **3**C. Structurally, the voltage regulator module **2** includes a circuit board assembly **20** and a magnetic core assembly **30**. The circuit board assembly **20** includes a printed circuit board **22**, at least one Dr.MOS element **21**, at least one first copper post **51**, at least one second copper post **52** and at least one third copper post **53**. The printed circuit board **22** has a first surface **22***a* and a second surface **22***b*, which are opposed to each other. The printed circuit board **22** further includes a lateral wall **22***c*, which is arranged between the first surface **22***a* and the second surface **22***b*. All Dr.MOS elements **21** and all input capacitors Cin are mounted on the first surface **22***a* of the printed circuit board **22** by a welding process or through a conductive adhesive (see FIG. **3**A). In addition, the input capacitors Cin are electrically connected with the Dr.MOS elements **21**. In this embodiment, the installation directions of the two Dr.MOS elements **21** on the first surface **22***a* of the printed circuit board **22** are opposed. Consequently, the input voltage pins of the two Dr.MOS elements **21** are arranged near each other. The input capacitors Cin are arranged between the two Dr.MOS elements **21**. That is, the input capacitors Cin are arranged between the input voltage pins of the two Dr.MOS elements **21**. The input capacitors Cin are shared by the two Dr.MOS elements **21**. Consequently, instead of routing wire, the nearby input voltage pins of the two Dr.MOS elements **21** are electrically connected with the input capacitors Cin, and the number of the input capacitors Cin is reduced.

[0046] The at least one first copper post **51**, the at least one second copper post **52** and the at least one third copper post **53** are mounted on the second surface **22***b* of the printed circuit board **22** by a welding process or through a conductive adhesive. A portion of a lateral surface of the second copper post **52** and a portion of a lateral surface of the third copper post **53** are coplanar with the lateral wall **22***c* of the printed circuit board **22**. The at least one first copper post **51** is used as the positive output terminal of the voltage regulator module **2**. The at least one second copper post **52** is used as the positive input terminal of the voltage regulator module **2**. The at least one third copper post **53** is used as the negative output terminal of the voltage regulator module **2**.

[0047] In an embodiment, the circuit board assembly **20** of the voltage regulator module **2** further includes at least one fourth copper post **54**. For example, as shown in FIGS. **3**B and **3**C, the circuit board assembly **20** further includes a plurality of fourth copper posts **54**. A portion of a lateral surface of the fourth copper posts **54** is coplanar with the lateral wall **22***c* of the printed circuit board **22**. The plurality of fourth copper posts **54** are mounted on the second surface **22***b* of the printed circuit board **22** by a welding process or through a conductive adhesive. The plurality of fourth copper posts **54** are used as signal terminals of the voltage regulator module **2**. When the voltage regulator module **2** is welded on a system board (not shown), the voltage regulator module **2** is in communication with the system board through the plurality of fourth copper posts **54**.

[0048] In an embodiment, the voltage regulator module **2** is welded on the system board through the at least one first copper post **51**, the at least one second copper post **52**, the at least one third copper post **53** and the at least one fourth copper post **54** of the circuit board assembly **20**. Consequently, the size of the voltage regulator module **2** is reduced.

[0049] The magnetic core assembly **30** is disposed on the second surface **22***b* of the printed circuit board **22**. Moreover, the magnetic core assembly **30** includes at least one ring-shaped core **31**. In this embodiment, the voltage regulator module **2** is a two-phase buck converter. In other words, the magnetic core assembly **30** includes two ring-shaped cores **31**. Each ring-shaped core **31** has a corresponding opening **30***b*. When the magnetic core assembly **30** is disposed on the second surface **22***b* of the printed circuit board **22**, the first copper posts **51** are penetrated through the corresponding openings **30***b*. Consequently, two inductors L are defined by the corresponding first copper posts **51** and the magnetic core assembly **30** collaboratively.

[0050] In an embodiment, the magnetic core assembly **30** further includes at least one air gap **30***c*. As shown in FIGS. **3**B and **3**C, the magnetic core assembly **30** includes two air gaps **30***c*. The two air gaps **30***c* are formed in the corresponding ring-shaped cores **31** respectively and located at two opposite sides of the magnetic core assembly **30**. In an embodiment, the two ring-shaped cores **31** of the magnetic core assembly **30** are integrally formed as an 8-shaped one-piece structure. Alternatively, the ring-shaped cores **31** are individual components and combined as the magnetic core assembly **30**.

[0051] In this embodiment, the opening **30***b* of the magnetic core assembly **30** has a circular shape. The first copper post **51** also has a circular shape. It is noted that the profiles of the opening **30***b* and the first copper post **51** are not restricted. For example, the opening **30***b* has a rectangular shape or any other appropriate shape, and the first copper post **51** has the shape matching the opening **30***b*. An external surface **30***d* of the magnetic core assembly **30** has a circular shape. Moreover, the shape of the external surface **30***d* of the magnetic core assembly **30** is not restricted.

[0052] In an embodiment, some or all of the fourth copper posts **54** are previously disposed on a lead frame. Consequently, the lead frame with the fourth copper posts **54** can be welded on the printed circuit board **22** more easily, and the fourth copper posts **54** can be well disposed and positioned. Similarly, the at least one first copper post **51**, the at least one second copper post **52** and the at least one third copper post **53** may be previously disposed on a lead frame so as to increase the installation efficiency. Hereinafter, an example of using the lead frame with the fourth copper posts

54 in the manufacturing process of the voltage regulator module **2** will be illustrated with reference to FIGS. **5** and **6**.

[0053] FIG. **5** schematically illustrates a lead frame with the fourth copper posts. FIG. **6** schematically illustrates the concept of using the lead frame as shown in FIG. **5** to produce a plurality of voltage regulator modules. As shown in FIG. **5**, the fourth copper posts **54** are previously disposed on a lead frame **55**. The lead frame **55** includes a frame body **56**. The fourth copper posts **54** are coupled to the inner periphery of the frame body **56**. Since the lead frame **55** with the fourth copper posts **54** can be welded on the printed circuit board **22** more easily, the efficiency of installing the fourth copper posts **54** is enhanced. The above concept may be expanded to manufacture a plurality of voltage regulator modules **2**. Please refer to FIG. **6**. In some embodiments, a connection printed circuit board with a plurality of printed circuit boards **22** is provided. After a plurality of magnetic core assemblies **30** are mounted on the corresponding printed circuit boards **22**, a plurality of semi-finished products **4***a* of the voltage regulator modules **2** are produced. Then, a plurality of lead frames **55** are disposed on the second surfaces **22***b* of the corresponding printed circuit boards **22**. Then, the fourth copper posts **54** of each lead frame **55** are welded on the second surface **22***b* of the printed circuit board **22** of the corresponding semi-finished product **4***a*. Since the fourth copper posts **54** are firstly fixed by the lead frame **55** and then welded on the corresponding printed circuit board **22**, the fourth copper posts **54** will not be shifted during the welding process or in the subsequent process. For example, the fourth copper posts **54** are not shifted in the packaging process. In an embodiment, the at least one first copper post **51**, the at least one second copper post **52** and the at least one third copper post **53** are disposed on the second surface **22***b* of the corresponding printed circuit board **22** before the lead frame **55** is welded on the corresponding printed circuit board **22**. In another embodiment, the at least one first copper post **51**, the at least one second copper post **52** and the at least one third copper post **53** are welded on the second surface **22***b* of the corresponding printed circuit board **22** after the lead frame **55** is welded on the corresponding printed circuit board **22**. Please refer to FIG. **6** again. After the plurality of semi-finished products **4***a* of the voltage regulator modules **2** are packaged, the junctions between the frame body **56** and the fourth copper posts **54** of each semi-finished product **4***a* are cut off and the junctions between the adjacent semi-finished products **4***a* of the connection printed circuit board are cut off. Consequently, a plurality of individual voltage regulator modules **2** are produced.

[0054] Please refer to FIGS. **7**A and **7**B. FIG. **7**A is a schematic perspective view illustrating a package structure of the voltage regulator module according to a first embodiment of the present disclosure. FIG. **7**B is a schematic perspective view illustrating the package structure of the voltage regulator module as shown in FIG. **7**A and taken along another viewpoint. The voltage regulator module **2** further includes a molding compound layer **60**. The first surface **22***a* of the printed circuit board **22**, the components on the first surface **22***a* of the printed circuit board **22** (e.g., the Dr.MOS elements **21** and the input capacitors Cin), the second surface **22***b* of the printed circuit board **22** and the components on the second surface **22***b* of the printed circuit board **22** (e.g., the copper posts **51**, **52**, **53** and **54**) are encapsulated by the molding compound layer **60** in a double-sided molding manner. The molding compound layer **60** includes a first molding side **60***a*, a second molding side **60***b* and a lateral molding wall **60***c*. The first molding side **60***a* of the molding compound layer **60** and the first surface **22***a* of the printed circuit board **22** are located at the same side of the printed circuit board **22**. The second molding side **60***b* of the molding compound layer **60** and the second surface **22***b* of the printed circuit board **22** are located at the same side of the printed circuit board **22**. The first molding side **60***a* and the second molding side **60***b* are opposed to each other. The lateral molding wall **60***c* of the molding compound layer **60** is arranged between the first molding side **60***a* and the second molding side **60***b*. In some embodiments, the first molding side **60***a* of the molding compound layer **60** is milled to adjust the thermal resistances of the heat generation components on the first surface **22***a* of the printed circuit board **22** (e.g., the Dr.MOS elements **21** and the input capacitors Cin). Optionally, after the first molding side **60***a* of the molding compound layer **60** is milled, the Dr.MOS elements **21** are exposed outside the first molding side **60***a* of the molding compound layer **60**. The external surface of the first molding side **60***a* of the molding compound layer **60** is a flat surface. The external surface is attached and fixed on a casing or a heat sink of the electronic device. Consequently, the thermal resistance is further reduced, and the heat dissipating capability of the voltage regulator module is increased.

[0055] Similarly, the second molding side **60***b* of the molding compound layer **60** is milled. Consequently, the end surfaces of the copper posts **51**, **52**, **53** and **54** are exposed outside the second molding side **60***b* of the molding compound layer **60** and coplanar with the second molding side **60***b* of the molding compound layer **60**. Moreover, the exposed portions of the copper posts **51**, **52**, **53** and **54** may be chemically plated. Consequently, the exposed portions of the copper posts **51**, **52**, **53** and **54** are used as bonding pads with larger areas.

[0056] Please refer to FIGS. **7**A and **7**B again. A first bonding pad **51***a* corresponding to the end surface of the first copper post **51** located in the second molding side **60***b* is used as a positive output terminal conductor of the voltage regulator module **2**. The first bonding pad **51***a* has a strip shape in the second molding side **60***b*. Moreover, the two ends of the first bonding pad **51***a* are connected with two lateral sides of the lateral molding wall **60***c* of the molding compound layer **60**. Moreover, two electroplated regions **60***d* are formed on the lateral sides of the lateral molding wall **60***c*. When the voltage regulator module **2** on the system board undergoes a reflow soldering process, the electroplated regions **60***d* can provide lateral wetting capability. Consequently, the welding strength of the combination between the voltage regulator module **2** and the system board is enhanced. Moreover, a second bonding pad **52***a* corresponding to the end surface of the second copper post **52** located in the second molding side **60***b* is used as a positive input terminal conductor of the voltage regulator module **2**. A third bonding pad **53***a* corresponding to the end surface of the third copper post **53** located in the second molding side **60***b* is used as a negative output terminal conductor of the voltage regulator module **2**. A fourth bonding pad **54***a* corresponding to the end surface of the fourth copper post **54** located in the second molding side **60***b* is used as a signal terminal conductor. The electroplated regions **60***d*, the second bonding pad **52***a*, the third bonding pad **53***a* and the fourth bonding pad **54***a* can increase the overall area of the bonding pad, reduce the average forces on the solder joints of the copper posts **51**, **52**, **53** and **54** and increase the strength of the solder joints. In addition, the weight withstood by the unit welding surface is reduced, the purpose of diffusing current is achieved, and the power loss is reduced. Moreover, since the first molding side **60***a* and the second molding side **60***b* of the molding compound layer **60** can be milled, the overall height of the voltage regulator module **2** is adjustable. Preferably, the area of the first bonding pad **51***a* (i.e., the positive output terminal conductor) is larger than or equal to the end surface area of the first copper post **51**, the area of the second bonding pad **52***a* (i.e., the positive input terminal conductor) is larger than or equal to the end surface area of the second copper post **52**, and the area of the third bonding pad **53***a* (i.e., the negative output terminal conductor) is larger than or equal to the end surface area of the third copper post **53**.

[0057] In an embodiment, the second molding side **60***b* of the molding compound layer **60** is milled such that a surface of the magnetic core assembly **30** is exposed. When the voltage regulator module **2** is welded on the system board, the exposed surface of the magnetic core assembly **30** can be attached on the system board directly. Consequently, the heat dissipating capability of the magnetic core assembly **30** is enhanced.

[0058] In some embodiments, the lateral molding wall **60***c* of the molding compound layer **60** is milled. Consequently, portions of the surfaces of the copper posts **52**, **53** and **54** are exposed outside the lateral molding wall **60***c* of the molding compound layer **60**. Moreover, the exposed portions of the surfaces of the copper posts **52**, **53** and **54** may be chemically plated. Consequently, the portion of the second copper post **52** exposed outside the lateral molding wall **60***c* is connected with the second bonding pad **52***a* (i.e., the positive input terminal conductor), the portion of the third copper post **53** exposed outside the lateral molding wall **60***c* is connected with the third bonding pad **53***a* (i.e., the negative output terminal conductor), and the portion of the fourth copper post **54** exposed outside the lateral molding wall **60***c* is connected with the fourth bonding pad **54***a* (i.e., the signal terminal conductor). When the voltage regulator module **2** on the system board undergoes a reflow soldering process, the electroplated regions of the copper posts **52**, **53** and **54** can provide lateral wetting capability. Consequently, the welding strength of the combination between the voltage regulator module **2** and the system board is enhanced.

[0059] Please refer to FIGS. **3**B and **3**C. The circuit board assembly **20** of the voltage regulator module **2** includes two first copper posts **51** and two second copper posts **52**. The two first copper posts **51** are located at a middle region of the printed circuit board **22**. The two second copper posts **52** are respectively located beside two sides of the midpoint of the two first copper posts **51**. The two second copper posts **52** are disposed on the second surface **22***b* of the printed circuit board **22** and symmetric with respect to the midpoint of the two first copper posts **51**. Moreover, the two second copper posts **52** are arranged beside the region between the two first copper posts **51**. The circuit board assembly **20** of the voltage regulator module **2** includes two third copper posts **53**. The second surface **22***b* of the printed circuit board **22** has four corners. Two corners are arranged along a first diagonal line and symmetric with respect to the midpoint of the second surface **22***b* of the printed circuit board **22**. The other two corners are arranged along a second diagonal line and symmetric with respect to the midpoint of the second surface **22***b* of the printed circuit board **22**. The two third copper posts **53** are disposed on the second surface **22***b* of the printed circuit board **22** and located at the two corners along the first diagonal line, respectively. Moreover, the two third copper posts **53** are respectively located beside two sides of the midpoint of the two first copper posts **51**. Portion of the plurality of fourth copper posts **54** and the rest of the plurality of fourth copper posts **54** are disposed on the second surface **22***b* of the printed circuit board **22** and located at the two corners along the second diagonal line respectively. When the voltage regulator module **2** is welded on the system board, the voltage regulator module **2** is in communication with the system board through the signal terminals.

[0060] In some embodiments, the magnetic core assembly **30** is fixed on the second surface **22***b* of the printed circuit board **22** through a thermal adhesive. Consequently, the heat dissipating capability of the magnetic core assembly **30** is enhanced. In this embodiments, there is a first gap **23** between the external surface **30***d* of the magnetic core assembly **30** and the second copper post **52** (and the third copper post **53**), and there is a second gap **24** between an inner wall **30***e* of the magnetic core assembly **30** and the first copper post **51**. An underfill material is filled in the first gap **23** and the second gap **24**. Consequently, the magnetic core assembly **30**, the printed circuit board **22** and the adjacent copper posts are contacted with each other through the underfill material. Due to the underfill material, no vacant spaces exit in the first gap **23** and the second gap **24** after the packaging process is completed.

[0061] As mentioned above, the voltage regulator module **2** is encapsulated by the molding compound layer **60** through the double-sided plastic molding process. Moreover, the copper posts **51**, **52**, **53** and **54** are chemically plated to generate the corresponding bonding pads. The bonding pads are electrically connected with the system board. Consequently, when the voltage regulator module on the system board undergoes a reflow soldering process, the inner components to be reheated are not detached or shifted. In other words, the process of fabricating the voltage regulator module **2** is simplified.

[0062] It is noted that the example of the voltage regulator module may be modified. For example, the magnetic core assembly **30**, the first copper posts **51**, the second copper posts **52** and the third copper posts **53** are embedded within the printed circuit board **22**. Moreover, the second surface **22***b* of the printed circuit board **22** is electroplated to form the first bonding pad **51***a* (i.e., the positive output terminal conductor), the second bonding pad **52***a* (i.e., the positive input terminal conductor), the third bonding pad **53***a* (i.e., the negative output terminal conductor) and the fourth bonding pad **54***a*.

[0063] FIG. **8** is a schematic perspective view illustrating a package structure of a voltage regulator module according to a second embodiment of the present disclosure. In comparison with the above embodiment, the printed circuit board **22** includes an inner layer **80**. That is, the printed circuit board **22** is a multilayered printed circuit board **22**. As shown in FIG. **3**B, the magnetic core assembly **30** has a top surface **30***f* and a bottom surface **30***g*. After an electroplating process is performed, a first copper electrode and a second copper electrode are formed on the top surface **30***f* and the bottom surface **30***g*, respectively. Moreover, the magnetic core assembly **30**, the first copper posts **51**, the second copper posts **52** and the third copper posts **53** are fixed and embedded within the inner layer **80** of the printed circuit board **22**.

[0064] FIG. **9** is a schematic top cross-sectional view illustrating the inner layer of the printed circuit board of the voltage regulator module as shown in FIG. **8**. The inner layer **80** includes a switch region **81**, a magnetic core region **82**, a positive input region **83** and a negative output region **84**. The first copper posts **51** are disposed and fixed on the switch region **81**. The magnetic core assembly **30** is disposed and fixed on the magnetic core region **82**. The second copper posts **52** are disposed and fixed on the positive input region **83** and used as the positive input terminal of the voltage regulator module **2**. The third copper posts **53** are disposed and fixed on the negative output region **84** and used as the negative output terminal of the voltage regulator module **2**. In some embodiments, the magnetic core region **82** is connected with the positive input region **83** to form a connection region. Alternatively, the magnetic core region **82** is connected with the negative output region **84** to form a connection region.

[0065] In an embodiment, a plurality of plated through-holes corresponding to the first copper posts **51**, the second copper posts **52**, the third copper posts **53**, the first copper electrode and the second copper electrode are formed in the first surface **22***a* and the second surface **22***b* of the printed circuit board **22**. The switch region **81**, the magnetic core region **82**, the positive input region **83** and the negative output region **84** are connected with corresponding pins of the Dr.MOS elements **21**, which are mounted on the first surface **22***a* of the printed circuit board **22** through the corresponding plated through-holes.

[0066] In some embodiments, the inner layer can be implemented in different types. For example, the copper posts are replaced by other types of the conductive portion. FIG. **10**A is a schematic perspective view illustrating a voltage regulator module according to a third embodiment of the present disclosure. FIG. **10**B is a schematic exploded view illustrating the voltage regulator module of FIG. **10**A. FIG. **10**C is a schematic cross-sectional view illustrating the inner layer of the printed circuit board of the voltage regulator module of FIG. **10**A. As shown in FIGS. **10**A to **10**C, the voltage regulator module **2***a* of this embodiment is similar to the voltage regulator module **2** of the first embodiment. In this embodiment, the printed circuit board **22** of the circuit board assembly **20** of the voltage regulator module **2***a* includes an upper surface **221**, a lower surface **222**, at least one upper layer **223**, an inner layer **80***a*, at least one lower layer **224**, and a plurality of conductive portions **225**. The upper surface **221** and the lower surface **222** of the printed circuit board **22** are opposite to each other.

[0067] The at least one upper layer **223** and the at least one lower layer **224** are disposed in two opposite sides of the printed circuit board **22**. The upper surface **221** of the printed circuit board **22** is formed by one surface of the at least one upper layer **223**. In this embodiment, the plurality of switch elements **21** are disposed on the upper surface **221** of the printed circuit board **22**. The lower surface **222** of the printed circuit board **22** is formed by one surface of the at least one lower layer **224**. And plurality of connection pads are disposed on the lower surface **222** of the printed circuit board **22**, for electrical connecting with an external circuit or system. The other surface of the at least one lower layer **224** faces to the other surface of the at least one upper layer **223**.

[0068] The inner layer **80***a* is disposed between the at least one upper layer **223** and the at least one lower layer **224**, and embedded within the printed circuit board **22**. In this embodiment, the inner layer **80***a* includes an accommodation space **801**. The accommodation space **801** is embedded within the inner layer **80***a*. The plurality of conductive portions **225** are electrically connected between the upper surface **221** and the lower surface **222** of the printed circuit board **22**. In this embodiment, the plurality of conductive portions **225** are plated holes. Some conductive portions **225** run through the upper surface **221** and the lower surface **222** of the printed circuit board **22**. Some conductive portions **225** may be buried vias buried in the printed circuit board **22** below the upper surface **221** or the lower surface **222**. For example, the conductive portions **225** may be formed in the inner conductive layers but not occupying area on the surface conductive layers.

[0069] The magnetic core assembly **30** is accommodated within the accommodation space **801** of the inner layer **80***a* and embedded within the printed circuit board **22**. In this embodiment, the materials of the inner layer **80***a* is cured and fully covers the magnetic core assembly **30**. The thickness of the inner layer **80***a* is larger than or equal to the height of the magnetic core assembly **30**.

[0070] In some embodiment, the at least one lower layer **224** may not be included, and the lower surface **222** of the printed circuit board **22** is formed by one surface of the inner layer **80***a*.

[0071] FIG. **11**A is a schematic perspective view illustrating a voltage regulator module according to a fourth embodiment of the present disclosure. FIG. **11**B is a schematic exploded view illustrating the voltage regulator module of FIG. **11**A. FIG. **11**C is a schematic cross-sectional view illustrating the printed circuit board of the voltage regulator module of FIG. **11**A. As shown in FIGS. **11**A to **11**C, the voltage regulator module **2***b* of this embodiment is similar to the voltage regulator module **2** of the first embodiment. In this embodiment, the printed circuit board **22** of the circuit board assembly **20** of the voltage regulator module **2***b* includes an upper surface **221**, a lower surface **222**, a first sub printed circuit board **223**, an inner layer **80***b* and a plurality of conductive portions **225**. In the embodiment, an independent circuit board (e.g. PCB) is used as the inner layer **80***b*. The inner layer **80***b* can be a single layer or an inner layer assembly including a plurality of conductive layers. The upper surface **221** and the lower surface **222** of the printed circuit board **22** are opposite to each other.

[0072] The upper surface **221** of the printed circuit board **22** is formed by one surface of the first sub printed circuit board **223**. In this embodiment, the plurality of switch elements **21** are disposed on the upper surface **221** of the printed circuit board **22**. The lower surface **222** of the printed circuit board **22** is formed by one surface of the inner layer **80***b*. And plurality of connection pads are disposed on the lower surface **222** of the printed circuit board **22**, for electrical connecting with an external circuit or system

[0073] The inner layer **80***b* is disposed adjacent to the first sub printed circuit board **223**, and embedded within the printed circuit board **22**. In this embodiment, the inner layer **80***b* includes an accommodation space **801**. The accommodation space **801** is concaved formed from a surface of the inner layer **80***b* and covered by the first sub printed circuit board **223**. In this embodiment, the first sub printed circuit board **223** is connected with the inner layer **80***b* through a welding means. The plurality of conductive portions **225** include at least portion of the plurality of conductive portions **225** in the inner layer **80***b* and another portion of the plurality of conductive portions **225** in the first sub printed circuit board **223**. The plurality of conductive portions **225** connect the upper surface **221** and the lower surface **222** of the printed circuit board **22**. The at least portion of the plurality of conductive portions **225** in the inner layer **80***b* or another portion of the plurality of the conductive portions **225** in the first sub printed circuit board **223** are plated holes. Some conductive portions in the inner layer **80***b* or some conductive portions in the first sub printed circuit board **223** run through a printed circuit board. Some conductive portions in the inner layer **80***b* or some conductive portions in the first sub printed circuit board **223** may be buried vias buried in the inner conductive layers of printed circuit board.

[0074] The magnetic core assembly **30** is accommodated within the accommodation space **801** of the inner layer **80***b* and embedded within the printed circuit board **22**. In this embodiment, the thickness of the inner layer **80***d* is larger than or equal to the height of the magnetic core assembly **30**. In this embodiment, a gap **802** is formed between the magnetic core assembly **30** and an inner surface of the accommodation space **801**.

[0075] FIGS. **11**D to **11**F are schematic cross-sectional views illustrating a process of manufacturing the printed circuit board of FIGS. **11**A to **11**C. As shown in FIG. **11**D, a first original printed circuit board **220***a* is provided. The first original printed circuit board **220***a* includes an accommodation space **801** and a plurality of original conductive portions **225***a*. The accommodation space **801** is concaved from a surface of the first original printed circuit board **220***a*. Some original conductive portions **225***a* run through and electrically connect two surfaces of the first original printed circuit board **220***a*. Some original conductive portions **225***a* may be buried vias buried in the inner conductive layers of the first original printed circuit board **220***a*. The first original printed circuit board **220***a* is used as the inner layer **80***b*. As shown in FIG. **11**E, the magnetic core assembly **30** is accommodated within the accommodation space **801** to form the inner layer **80***b*. As shown in FIG. **11**F, a second original printed circuit board **220***b* is disposed on the first original printed circuit board **220***a*. The accommodation space **801** is covered by the second original printed circuit board **220***b*. In this embodiment, the second original printed circuit board **220***b* includes at least one pad. The pad of the second original printed circuit board **220***b* and the corresponding original conductive portions **225***a* are connected to each other. The printed circuit board **22** is formed as FIG. **11**C.

[0076] FIG. **12**A is a schematic perspective view illustrating a voltage regulator module according to a fifth embodiment of the present disclosure. FIG. **12**B is a schematic exploded view illustrating the voltage regulator module of FIG. **12**A. FIG. **12**C is a schematic cross-sectional view illustrating the printed circuit board of the voltage regulator module of FIG. **12**A. As shown in FIGS. **12**A to **12**C, the voltage regulator module **2***c* of this embodiment is similar to the voltage regulator module **2** of the first embodiment. In this embodiment, the printed circuit board **22** of the circuit board assembly **20** of the voltage regulator module **2***c* includes an upper surface **221**, a lower surface **222**, at least one upper layer **223**, an inner layer **80***c*, at least one lower layer **224** and a plurality of conductive portions **225**. The upper surface **221** and the lower surface **222** of the printed circuit board **22** are opposite to each other.

[0077] The at least one upper layer **223** and the at least one lower layer **224** are disposed in two opposite sides of the printed circuit board **22**. The upper surface **221** of the printed circuit board **22** is formed by one surface of the at least one upper layer **223**. In this embodiment, the plurality of switch elements **21** are disposed on the upper surface **221** of the printed circuit board **22**. The lower surface **222** of the printed circuit board **22** is formed by one surface of the at least one lower layer **224**. Plurality of connection pads are disposed on the lower surface **222** of the printed circuit board **22**, for electrical connecting with an external circuit or system. The other surface of the at least one lower layer **224** faces to the other surface of the at least one upper layer **223**.

[0078] The inner layer **80***c* is composed of magnetic material layer and disposed between the at least one upper layer **223** and the at least one lower layer **224**, and embedded within the printed circuit board **22**. The plurality of conductive portions **225** are connected between the upper surface **221** and the lower surface **222** of the printed circuit board **22**. The conductive portions **225** are plated holes. Some conductive portions **225** run through the upper surface **221** and the lower surface **222** of the printed circuit board **22**. Some conductive portions **225** may be buried vias buried in the printed circuit board **22** below the upper surface **221** or the lower surface **222**. In this embodiment, some conductive portions **225** pass through the magnetic material layer.

[0079] FIG. **13**A is a schematic perspective view illustrating a voltage regulator module according to a sixth embodiment of the present disclosure. FIG. **13**B is a schematic exploded view illustrating the voltage regulator module of FIG. **13**A. As shown in FIGS. **13**A to **13**B, the voltage regulator module **2***d* of this embodiment is similar to the voltage regulator module **2** of the first embodiment. In this embodiment, the printed circuit board **22** of the circuit board assembly **20** of the voltage regulator module **2***d* includes an upper surface **221**, a lower surface **222**, at least one upper layer **223**, an inner layer **80***d*, at least one lower layer **224** and a plurality of conductive portions **225**. The upper surface **221** and the lower surface **222** of the printed circuit board **22** are opposite to each other.

[0080] The at least one upper layer **223** and the at least one lower layer **224** are disposed in two opposite sides of the printed circuit board **22**. The upper surface **221** of the printed circuit board **22** is formed by one surface of the at least one upper layer **223**. In this embodiment, the plurality of switch elements **21** are disposed on the upper surface **221** of the printed circuit board **22**. The lower surface **222** of the printed circuit board **22** is formed by one surface of the at least one lower layer **224**. And plurality of connection pads are disposed on the lower surface **222** of the printed circuit board **22**, for electrical connecting with an external circuit or system. The other surface of the at least one lower layer **224** faces to the other surface of the at least one upper layer **223**.

[0081] The inner layer **80***d* is disposed between the at least one upper layer **223** and the at least one lower layer **224**, and embedded within the printed circuit board **22**. In the embodiment, an independent circuit board (e.g. PCB) is used as the inner layer **80***d*. The inner layer **80***d* can be a single layer or an inner layer assembly including a plurality of conductive layers. In this embodiment, the inner layer **80***d* includes an accommodation space **801**. The accommodation space **801** is concaved formed from a surface of the inner layer **80***d* and covered by the first sub printed circuit board **223**. The plurality of conductive portions **225** are connected between the upper surface **221** and the lower surface **222** of the printed circuit board **22**. The conductive portions **225** are plated holes. Some conductive portions **225** run through the upper surface **221** and the lower surface **222** of the printed circuit board **22**. Some conductive portions **225** may be buried vias buried in the printed circuit board **22** below the upper surface **221** or the lower surface **222**.

[0082] The magnetic core assembly **30** is composed of liquid magnetic material layer and fully accommodated within the accommodation space **801** of the inner layer **80***d* and embedded within the printed circuit board **22**. In this embodiment, the thickness of the inner layer **80***d* is larger than and equal to the height of the magnetic core assembly **30**. In this embodiment, the at least one upper layer **233** and the at least one lower layer **224** are bonded with the inner layer **80***d* through a compression means.

[0083] In some embodiment, the at least one lower layer **224** may not be included, and the lower surface **222** of the printed circuit board **22** is formed by one surface of the inner layer **80***d*.

[0084] FIGS. **13**C to **13**F are schematic cross-sectional views illustrating a process of manufacturing the printed circuit board of FIGS. **13**A to **13**B. As shown in FIG. **13**C, a first original printed circuit board **220***a* is provided. The first original printed circuit board **220***a* includes an accommodation space **801**. The accommodation space **801** is concaved from a surface of the first original printed circuit board **220***a*. As shown in FIG. **13**D, the magnetic core assembly **30** is composed of liquid magnetic material layer and fully accommodated within the accommodation space **801**. The first original printed circuit board **220***a* is used as the inner layer **80***d*. As shown in FIG. **13**E, the at least one upper layer **233** and the second sub printed circuit board **224** are provided. The inner layer **80***d* is disposed between the at least one upper layer **233** and the second sub printed circuit board **224**. The accommodation space **801** is covered by the at least one upper layer **233**. In this embodiment, the at least one upper layer **233** and the second sub printed circuit board **224** are bonded with the inner layer **80***d* through a compression means. The printed circuit board **22** is formed. As shown in FIG. **13**F, the plurality of conductive portions **225** is formed in the printed circuit board **22**.

[0085] FIG. **14**A is a schematic cross-sectional view illustrating the package structure of the voltage regulator module as shown in FIG. **8**. The printed circuit board **22** includes first plated through-holes **91** and second plated through-holes (PTH) **92**. The first copper posts **51** and the corresponding pins of the Dr.MOS elements **21** on the first surface **22***a* of the printed circuit board **22** are electrically connected with each other through the corresponding first plated through-holes **91**. The second copper electrode of the magnetic core assembly **30** and the corresponding pins of the Dr.MOS elements **21** on the first surface **22***a* of the printed circuit board **22** are electrically connected with each other through the corresponding second plated through-holes **92**. Moreover, the printed circuit board **22** includes signal plated through-holes (not shown) for connecting the fourth bonding pad **54***a* with the corresponding pins of the Dr.MOS elements **21** on the first surface **22***a* of the printed circuit board **22**.

[0086] As shown in FIG. **14**A, the printed circuit board **22** further includes third plated through-holes **93** and fourth plated through-holes **94**. The first bonding pad **51***a* and the first copper electrode of the magnetic core assembly **30** are electrically connected with each other through the corresponding third plated through-holes **93**. The first bonding pad **51***a* and the first copper posts **51** are electrically connected with each other through the corresponding fourth plated through-holes **94**.

[0087] In this embodiment, the plated through-holes **91**, **92**, **93** and **94** are straight holes. It is noted that numerous modifications and alterations may be made while retaining the teachings of the disclosure. FIG. **14**B is a schematic cross-sectional view illustrating a variant example of the package structure of the voltage regulator module as shown in FIG. **14**A. In the variant example, the plated through-holes **91**, **92**, **93** and **94** are stepped holes.

[0088] Since the copper electrodes are disposed on the top surface and the bottom surface of the magnetic core assembly **30** and the copper electrodes are electrically connected with the bonding pads through the corresponding plated through-holes, the interface thermal resistance between the magnetic core assembly **30** and the surfaces of the printed circuit board **22** is reduced. In such way, the heat generated by the magnetic core assembly **30** can be transferred to the surfaces of the printed circuit board **22** through the corresponding plated through-holes and dissipated away.

[0089] Similarly, the second bonding pad **52***a* (i.e., the positive input terminal conductor) is electrically connected with the second copper posts **52** through corresponding plated through-holes (not shown), and the third bonding pad **53***a* (i.e., the negative output terminal conductor) is electrically connected with the third copper posts **53** through corresponding plated through-holes (not shown). In some embodiments, the fourth copper posts are replaced by corresponding signal through-holes (not shown). Moreover, one end of each signal through-hole is contacted with the corresponding pin of the Dr.MOS element **21**, and the other end of each signal through-hole is contacted with the fourth bonding pad **54***a*.

[0090] Moreover, the copper posts **52**, **53** and **54** are exposed to the lateral molding wall **60***c* of the molding compound layer **60**. The exposed portions of the copper posts **52**, **53** and **54** are electroplated to form electroplated regions. When the voltage regulator module **2** on the system board undergoes a reflow soldering process, the electroplated regions can provide lateral wetting capability. Consequently, the welding strength of the combination between the voltage regulator module **2** and the system board is enhanced.

[0091] In an embodiment, the height of the magnetic core assembly **30**, the height of the first copper post **51**, the height of the second copper post **52** and the height of the third copper post **53** are equal. Moreover, the magnetic core assembly **30**, the first copper post **51**, the second copper post **52** and the third copper post **53** are welded on the same inner layer **80** of the printed circuit board **22**. It is noted that the heights of the magnetic core assembly and the copper posts may be different from each other and the magnetic core assembly and the copper posts may be fixed on different inner layers.

[0092] In this embodiment, the magnetic core assembly **30** is disposed on the inner layer **80** of the printed circuit board **22**. Consequently, the second surface **22***b* of the printed circuit board **22** has more layout area for arranging the first bonding pad **51***a*, the second bonding pad **52***a*, the third bonding pad **53***a* and the fourth bonding pad **54***a*. Since the area of at least one of the first bonding pad **51***a*, the second bonding pad **52***a*, the third bonding pad **53***a* and the fourth bonding pad **54***a* is increased, the welding area and the welding strength between the corresponding bonding pad and the system board are increased and the average force on the voltage regulator module is reduced. In addition, the weight withstood by the unit welding surface is reduced, the purpose of diffusing current is achieved, and the power loss is reduced.

[0093] From the above descriptions, the present disclosure provides the voltage regulator module. The switch elements and the magnetic core assembly are disposed on two opposite sides of the printed circuit board. The first surface of the printed circuit board is encapsulated. The magnetic core assembly and the copper posts are embedded within the printed circuit board. The conductors formed on the second surface of the printed circuit board are welded on the system board. Alternatively, the copper posts are disposed on the circuit board, encapsulated by a double-sided plastic molding process, and welded on the system board. When the voltage regulator module on the system board undergoes a reflow soldering process, the inner components to be heated are not detached or shifted. In other words, the process of fabricating the voltage regulator module is simplified. When compared with the conventional voltage regulator module, the size of the voltage regulator module of the present disclosure is reduced.

[0094] While the disclosure has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is to be understood that the disclosure needs not be limited to the disclosed embodiment. On the contrary, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

What is claimed is:

1. A voltage regulator module, comprising:

a circuit board assembly comprising a printed circuit board and at least one switch element, wherein the printed circuit board comprises an upper surface, a lower surface, an inner layer and a plurality of conductive portions, the upper surface and the lower surface are opposite to each other, the inner layer is embedded within the printed circuit board, the plurality of conductive portions are electrically connected between the upper surface and the lower surface of the printed circuit board, and the at least one switch element is disposed on the upper surface of the printed circuit board; and

a magnetic core assembly disposed in the inner layer.

2. The voltage regulator module according to claim **1**, wherein at least one of the plurality of conductive portions is a plated hole.

3. The voltage regulator module according to claim **1**, wherein a thickness of the inner layer is larger than or equal to a height of the magnetic core assembly.

4. The voltage regulator module according to claim **1**, wherein the inner layer comprises at least one accommodation space, the at least one accommodation space is embedded within the inner layer, wherein the magnetic core assembly is accommodated within the at least one accommodation space.

5. The voltage regulator module according to claim **4**, wherein the printed circuit board further comprises at least one upper layer, wherein the upper surface of the printed circuit board is formed by one surface of the at least one upper layer, and the inner layer is disposed on another surface of the at least one upper layer and the at least one accommodation space is covered by the at least one upper layer.

6. The voltage regulator module according to claim **5**, wherein the printed circuit board comprises at least one lower layer, wherein the lower surface of the printed circuit board is formed by one surface of the at least one lower layer, and the inner layer is disposed between the at least one upper layer and the at least one lower layer.

7. The voltage regulator module according to claim **4**, wherein the printed circuit board further comprises a first sub printed circuit board, wherein the upper surface of the printed circuit board is formed by one surface of the first sub printed circuit board, and the inner layer is disposed on another surface of the first sub printed circuit board, and the at least one accommodation space is covered by the first sub printed circuit board.

8. The voltage regulator module according to claim **7**, wherein an original printed circuit board is used to form the inner layer.

9. The voltage regulator module according to claim **1**, wherein the inner layer is composed of magnetic material layer.

10. The voltage regulator module according to claim **9**, wherein the printed circuit board further comprises at least one upper layer and at least one lower layer, wherein the upper surface of the printed circuit board is formed by one surface of the at least one upper layer, the lower surface of the printed circuit board is formed by one surface of the at least one lower layer, and the magnetic material layer is disposed between the at least one upper layer and the at least one lower layer.

11. The voltage regulator module according to claim **9**, wherein at least one of the plurality of conductive portions passes through the magnetic material layer.

12. The voltage regulator module according to claim **1**, wherein the circuit board assembly comprises a first bonding pad, a second bonding pad, a third bonding pad and a fourth bonding pad, the first bonding pad is disposed on the lower surface of the printed circuit board and electrically connected with at least corresponding one of the plurality of conductive portions to form a positive output terminal conductor of the voltage regulator module, the second bonding pad is disposed on the lower surface of the printed circuit board and electrically connected with at least corresponding one of the plurality of conductive portions to form a positive input terminal conductor of the voltage regulator module, the third bonding pad is disposed on the lower surface of the printed circuit board and electrically connected with at least corresponding one of the plurality of conductive portions to form a negative output terminal conductor of the voltage regulator module, the fourth bonding pad is disposed on the lower surface of the printed circuit board and electrically connected with at least corresponding at least one of the plurality of conductive portions to form a signal terminal conductor of the voltage regulator module.

13. The voltage regulator module according to claim **1**, wherein at least one of the plurality of conductive portions passes through the magnetic core assembly, so that at least one inductor is defined by the at least one conductive portion and the magnetic core assembly collaboratively.

14. The voltage regulator module according to claim **13**, wherein the magnetic core assembly comprises at least one opening, wherein at least one of the plurality of conductive portions passes through corresponding one of the at least one opening.

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